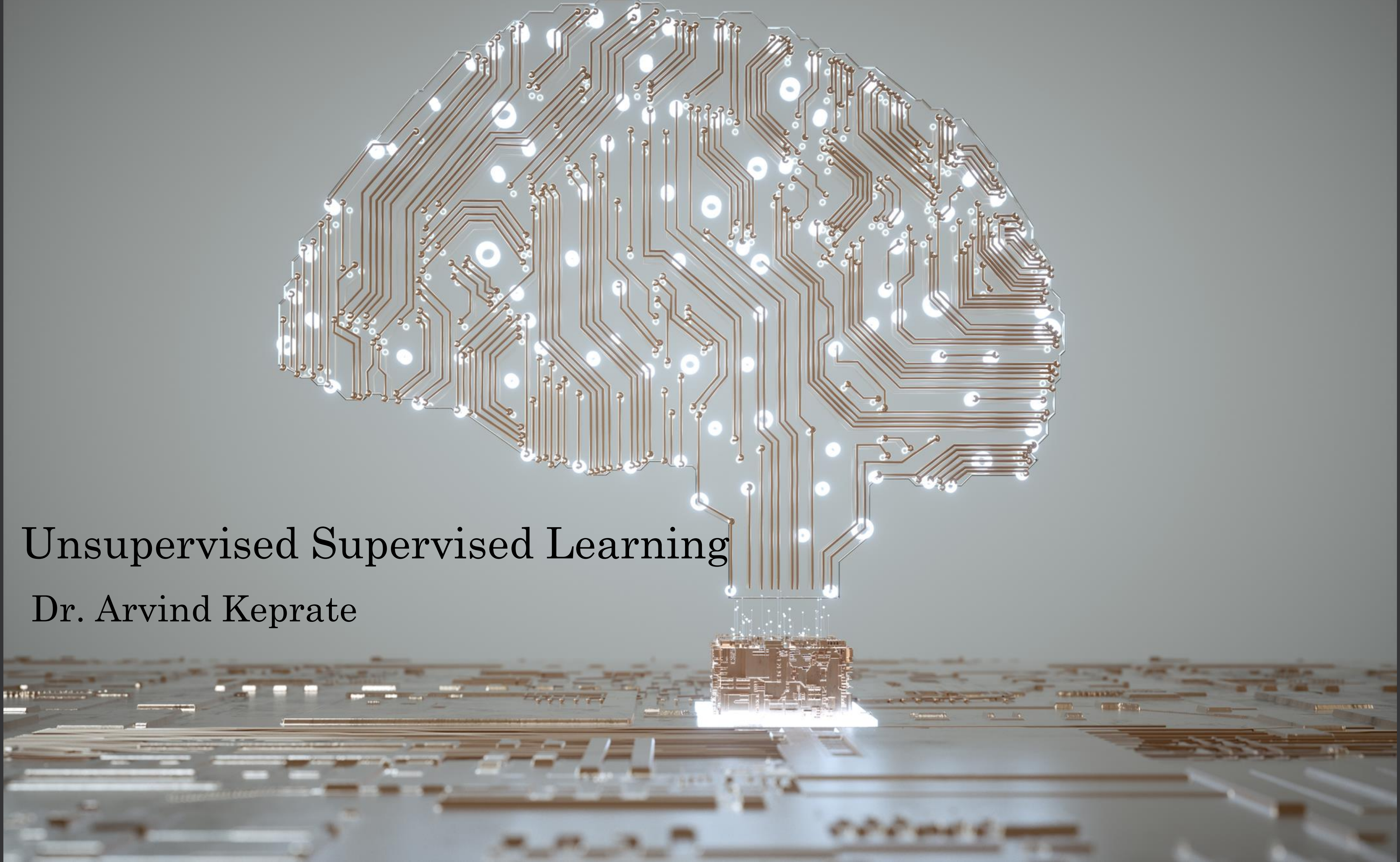
ML & NLP: PGR210



Unsupervised Supervised Learning

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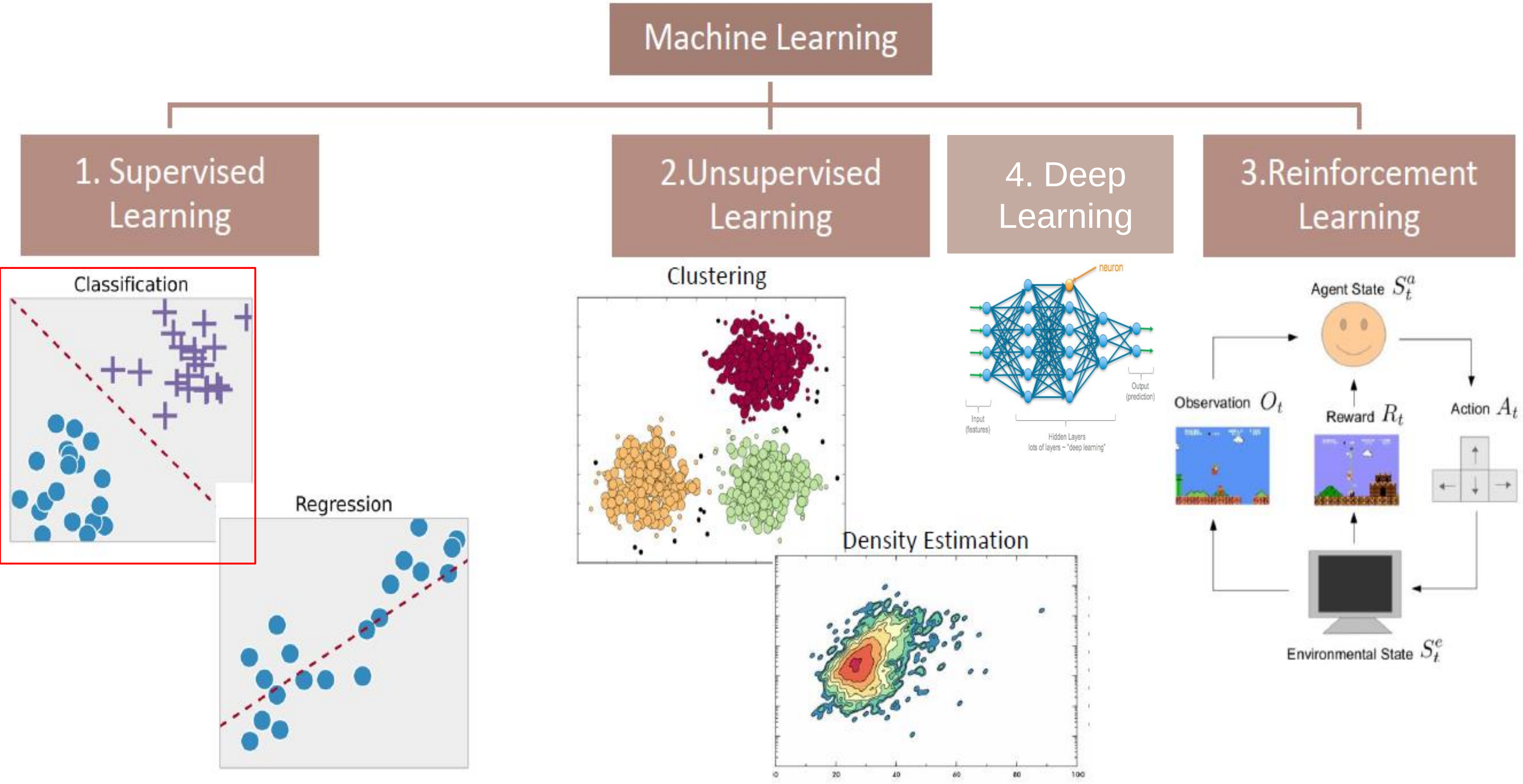
Topics for Today

- Supervised vs. UnSupervised ML
- Types of Unsupervised ML
- What is Clustering?
- k-Means Clustering
- Heirarchial Clustering
- Density Based Clustering
- Clustering Applications

Types of ML?



Machine Learning
1. Supervised Learning
2.Unsupervised Learning
4. Deep Learning
3.Reinforcement Learning
Classification
Regression
Clustering
Density Estimation
neuron
Input (features)
Hidden Layers
lots of layers ~ "deep learning"
Output (prediction)
Agent State S_t^a
Observation O_t
Reward R_t
Action A_t
Environmental State S_t^e

Supervised vs Unsupervised ML

	Supervised	Unsupervised
Data	(x, y) where x is data, y is label	x - just data, no labels
Goal	Learn a function to map $x \rightarrow y$ Form a model for $\mathrm{P}(x\|y)$, where y is the label for x	Learn underlying hidden structure of the data Form a model for $\mathrm{P}(x)$, where x is an input vector
Methods	Regression, object detection, semantic segmentation, image captioning, etc.	Clustering, dimensionality reduction, feature learning, density estimation, etc.
Example		

Classical Machine Learning

Task Driven

Data Driven

Supervised Learning

(Pre Categorized Data)

Classification

(Divide the socks by Color)

Eg. Identity Fraud Detection

Regression

(Divide the Ties by Length)

Eg. Market Forecasting

Unsupervised Learning

(Unlabelled Data)

Clustering

(Divide by Similarity)

Eg. Targeted Marketing

Association

(Identify Sequences)

Eg. Customer Recommendation

Dimensionality Reduction

(Wider Dependencies)

Eg. Big Data Visualization

Obj: Predications & Predictive Models

Pattern/ Structure Recognition

Association

- An association rule is an unsupervised learning method that is used for finding the relationships between variables in the large database. It determines the set of items that occurs together in the dataset.

The Parable of Diapers and Beers

Some time ago, Wal-Mart decided to combine the data from its loyalty card system with that from its point of sale systems. The former provided Wal-Mart with demographic data about its customers, the latter told it where, when and what those customers bought. Once combined, the data was mined extensively and many correlations appeared. Some of these were obvious; people who buy gin are also likely to buy tonic. They often also buy lemons. However, one correlation stood out like a sore thumb because it was so unexpected.

On Friday afternoons, young American males who buy diapers (nappies) also have a predisposition to buy beer. No one had predicted that result, so no one would ever have even asked the question in the first place.

Market Basket Analysis

ID	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}
…	…

market basket transactions

{Diapers, Beer} Example of a frequent itemset

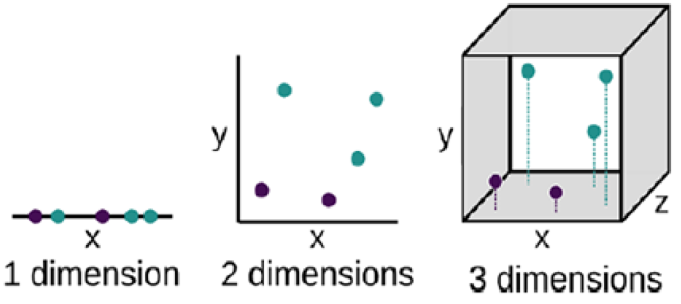
{Diapers} → {Beer} Example of an association rule

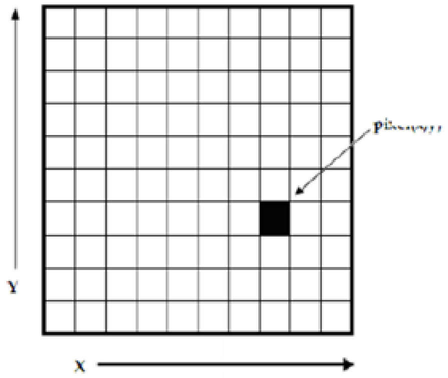
Market Basket Analysis

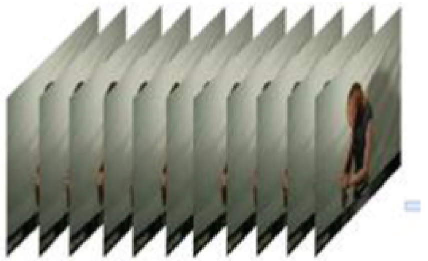
What did the supermarket do with this knowledge?

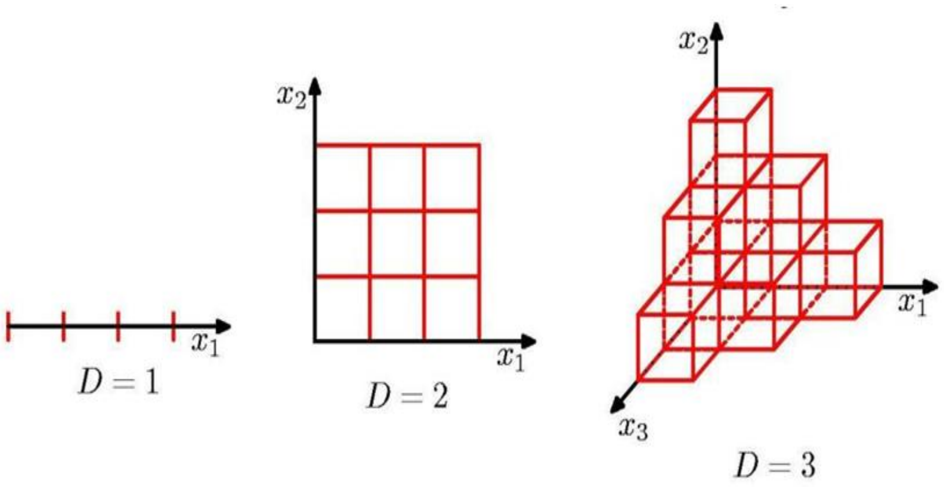
- They put the premium beer display next to the diapers
- The result was that the fathers buying diapers and who also usually bought beer now bought the premium beer (the up-sell) as it was so conveniently placed next to the diapers
- Significantly, the men that did not buy beer before began to purchase it because it was so visible and handy - just next to the nappies (the cross-sell).
- Beer sales skyrocketed

Dimensionality Reduction









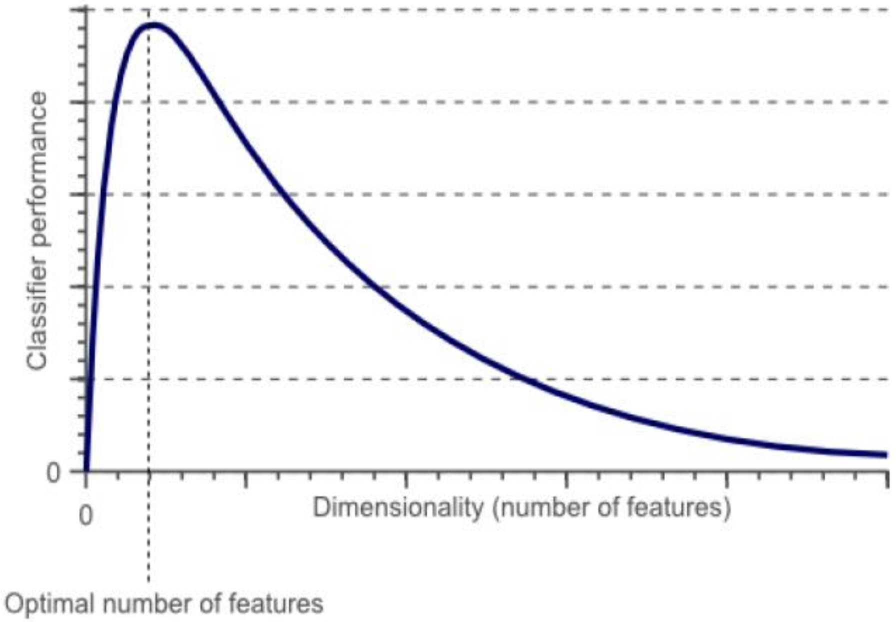
- No. of cells grow exponentially with D
- Need exponentially large no. of training data points
- Not a good approach for more than a few dimensions!

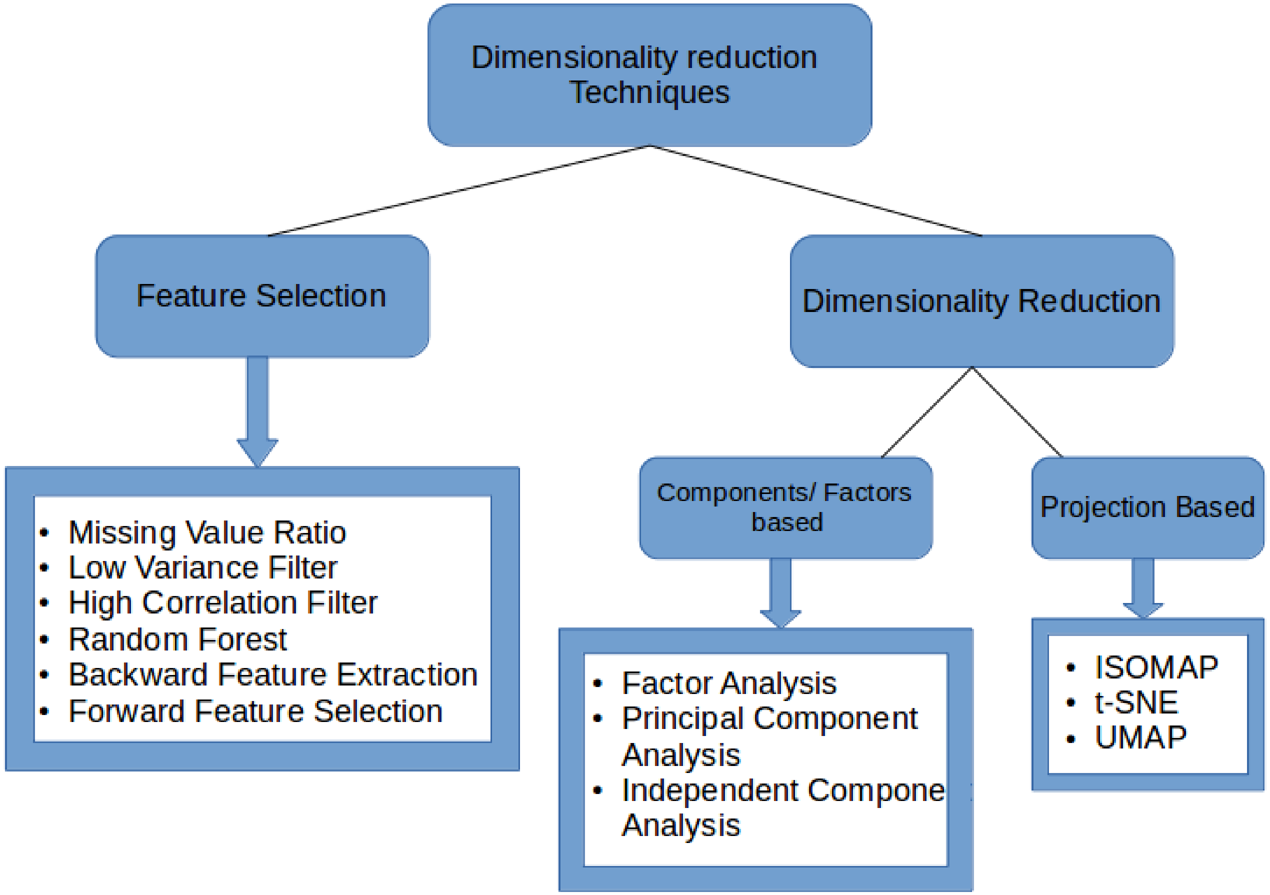
Reference: Christopher M Bishop: Pattern Recognition & Machine Leaning, 2006 Springer

- As the number of features (dimensions) grows, the amount of data we need to generalize grows exponentially.

Dimensionality Reduction

- Dimensionality Reduction is the transformation of data from a high-dimensional space into a low-dimensional space so that the low-dimensional representation retains some meaningful properties of the original data, ideally close to its intrinsic dimension.

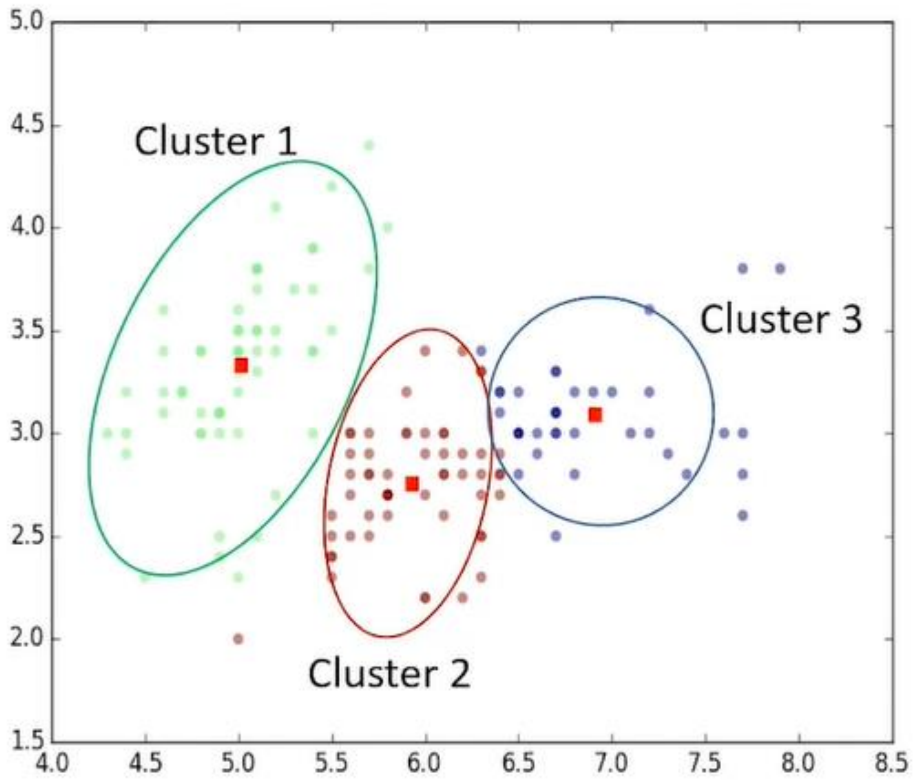




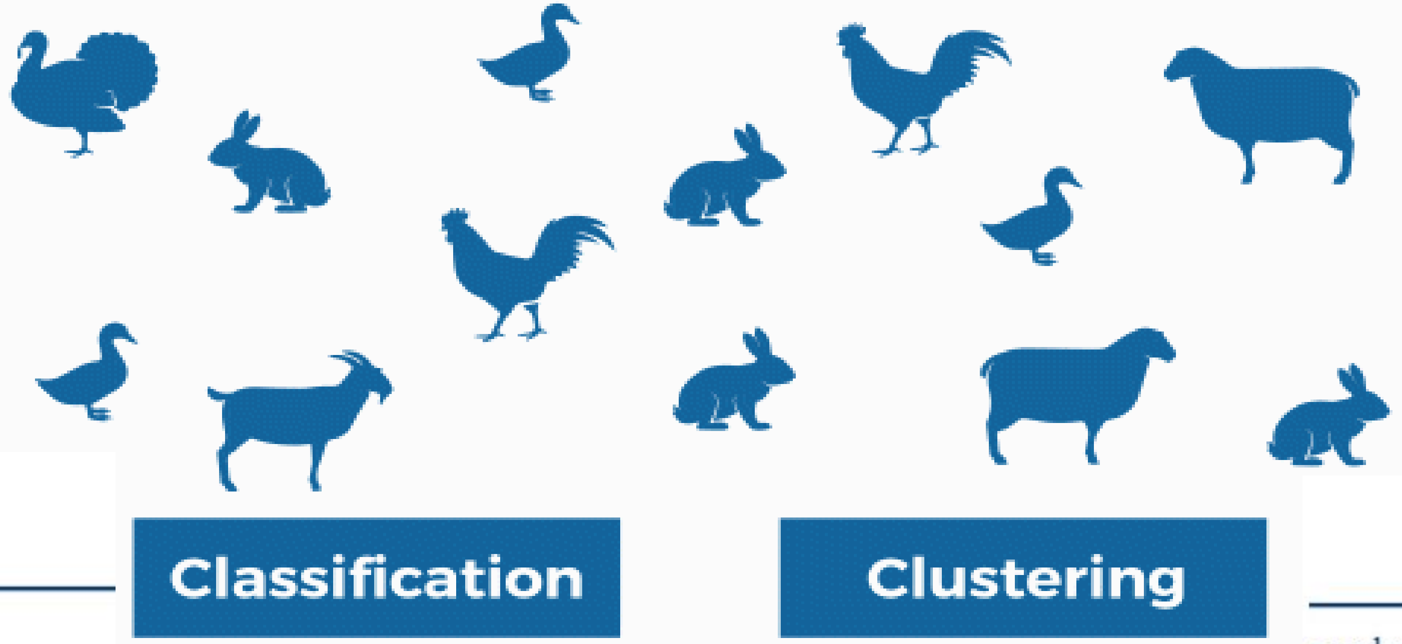
What is Clustering?

What is a cluster?

A group of objects that are **similar to other objects** in the cluster, and **dissimilar to data points** in other clusters.



Classification vs. Clustering



Classification

- known number of classes
- based on a training set
- used to classify future observations
- Classification is a form of supervised learning

Clustering

- unknown number of classes
- no prior knowledge
- used to understand (explore) data
- Clustering a form of unsupervised learning

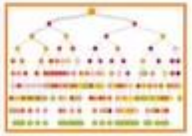
Classification vs. Clustering

Labeled dataset

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio	Defaulted
1	41	2	6	19	0.124	1.073	NBA001	6.3	0
2	47	1	26	100	4.582	8.218	NBA021	12.8	0
3	33	2	10	57	6.111	5.802	NBA013	20.9	1
4	29	2	4	19	0.681	0.516	NBA009	6.3	0
5	47	1	31	253	9.308	8.908	NBA008	7.2	0
6	40	1	23	81	0.998	7.831	NBA016	10.9	1
7	38	2	4	56	0.442	0.454	NBA013	1.6	0
8	42	3	0	64	0.279	3.945	NBA009	6.6	0
9	26	1	5	18	0.575	2.215	NBA006	15.5	1
10	47	3	23	115	0.653	3.947	NBA011	4	0

Modeling

Prediction



Decision Tree

Unlabeled dataset

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio	Defaulted
1	41	2	6	19	0.124	1.073	NBA001	6.3	0
2	47	1	26	100	4.582	8.218	NBA021	12.8	0
3	33	2	10	57	6.111	5.802	NBA013	20.9	1
4	29	2	4	19	0.681	0.516	NBA009	6.3	0
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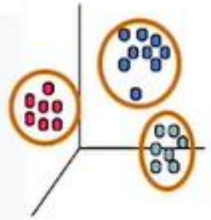
8	42	3	0	64	0.279	3.945	NBA009	6.6	0
9	26	1	5	18	0.575	2.215	NBA006	15.5	1

Modeling

Segmentation

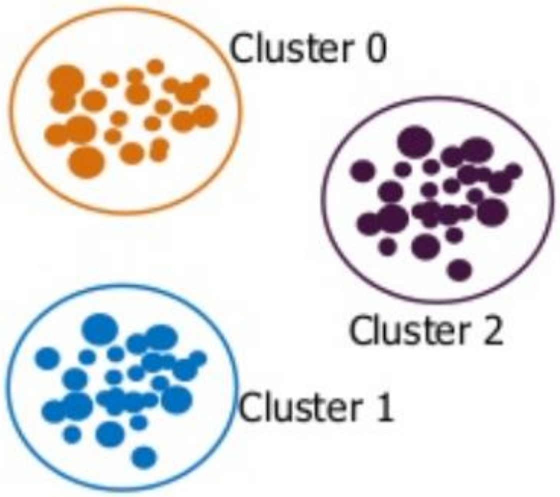


K-Means

Clustering Types

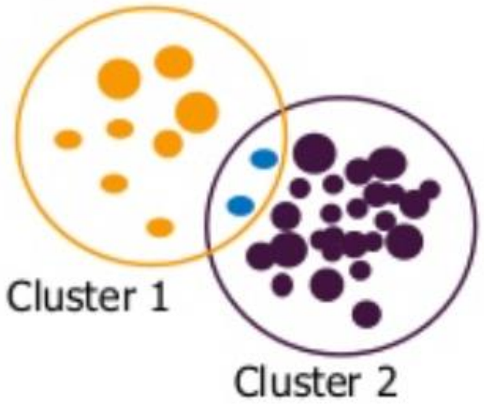
Exclusive Clustering

- An item belongs exclusively to one cluster, not several.
- K-means does this sort of exclusive clustering.



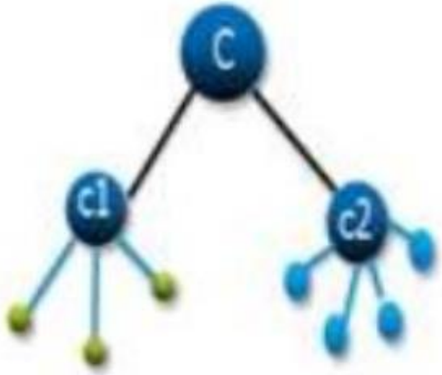
Overlapping Clustering

- An item can belong to multiple clusters
- Its degree of association with each cluster is known
- Fuzzy/ C-means does this sort of exclusive clustering.



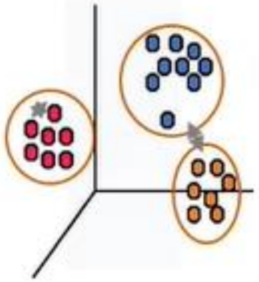
Hierarchial Clustering

- When two cluster have a parent-child relationship or a tree-like structure then it is Hierarchical clustering

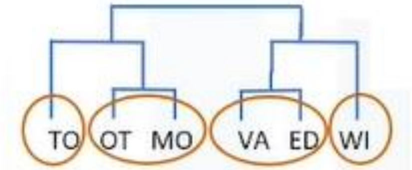


Clustering Algorithms

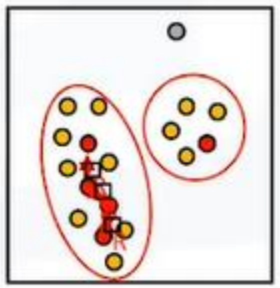
- Partitioned-based Clustering
  - Relatively efficient
  - E.g. k-Means, k-Median, Fuzzy c-Means



- Hierarchical Clustering
  - Produces trees of clusters
  - E.g. Agglomerative, Divisive



- Density-based Clustering
  - Produces arbitrary shaped clusters
  - E.g. DBSCAN



More Details: https://neptune.ai/blog/clustering-algorithms

Clustering for Segmentation

Customer Id	Age	Edu	Years Employed	Income	Card Debt	Other Debt	Address	DebtIncomeRatio	Defaulted
1	41	2	6	19	0.124	1.073	NBA001	6.3	0
2	47	1	26	100	4.582	8.218	NBA021	12.8	0
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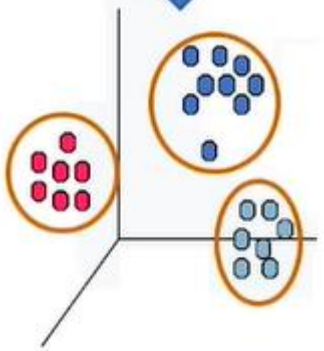
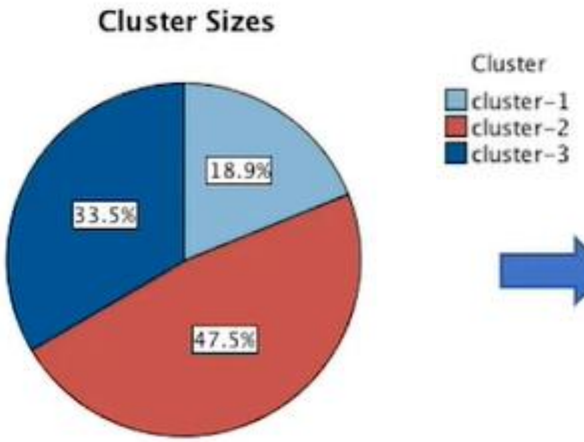
8	42	3	0	64	0.279	3.945	NBA009	6.6	0
9	26	1	5	18	0.575	2.215	NBA006	15.5	1

Customer ID	Segment
1	YOUNG AND LOW INCOME
2	AFFULUENT AND MIDDLE AGED
3	AFFULUENT AND MIDDLE AGED
4	YOUNG AND LOW INCOME
5	AFFULUENT AND MIDDLE AGED
6	AFFULUENT AND MIDDLE AGED
7	YOUNG AND LOW INCOME
8	YOUNG AND LOW INCOME
9	AFFULUENT AND MIDDLE AGED



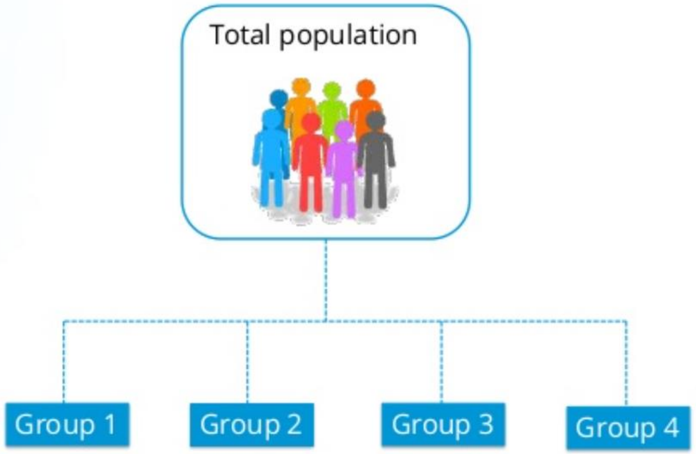


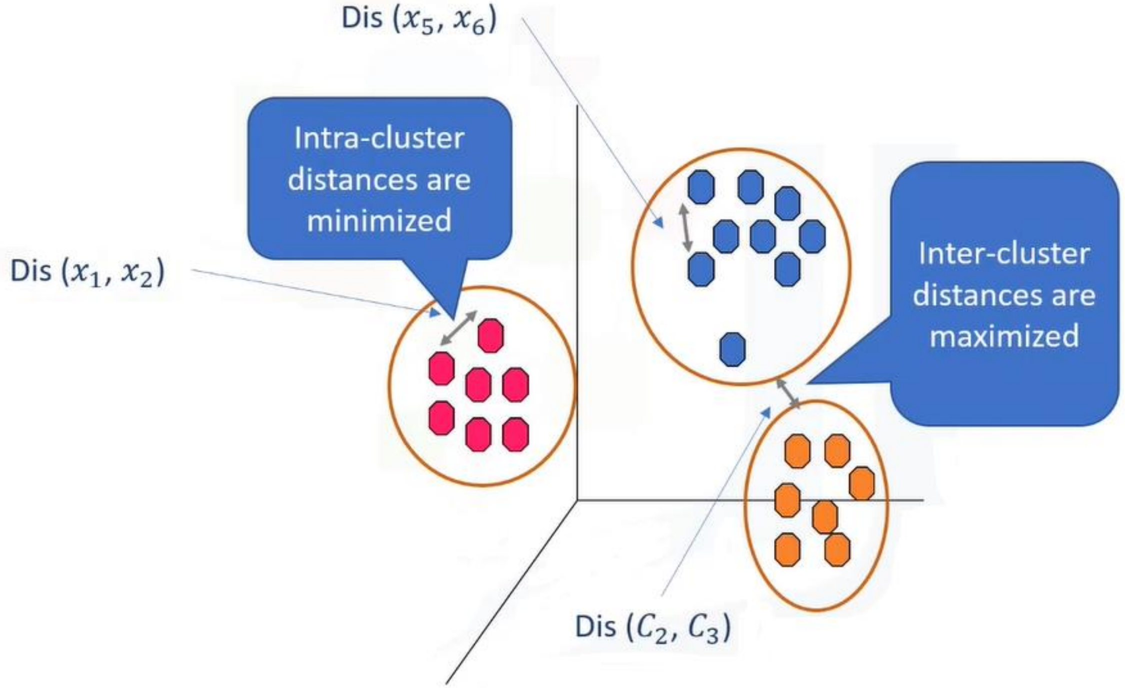
Cluster	Segment Name
cluster-1	AFFULUENT AND MIDDLE AGED
cluster-2	YOUNG EDUCATED AND MIDDLE INCOME
cluster-3	YOUNG AND LOW INCOME

K-Means Clustering

k-means clustering

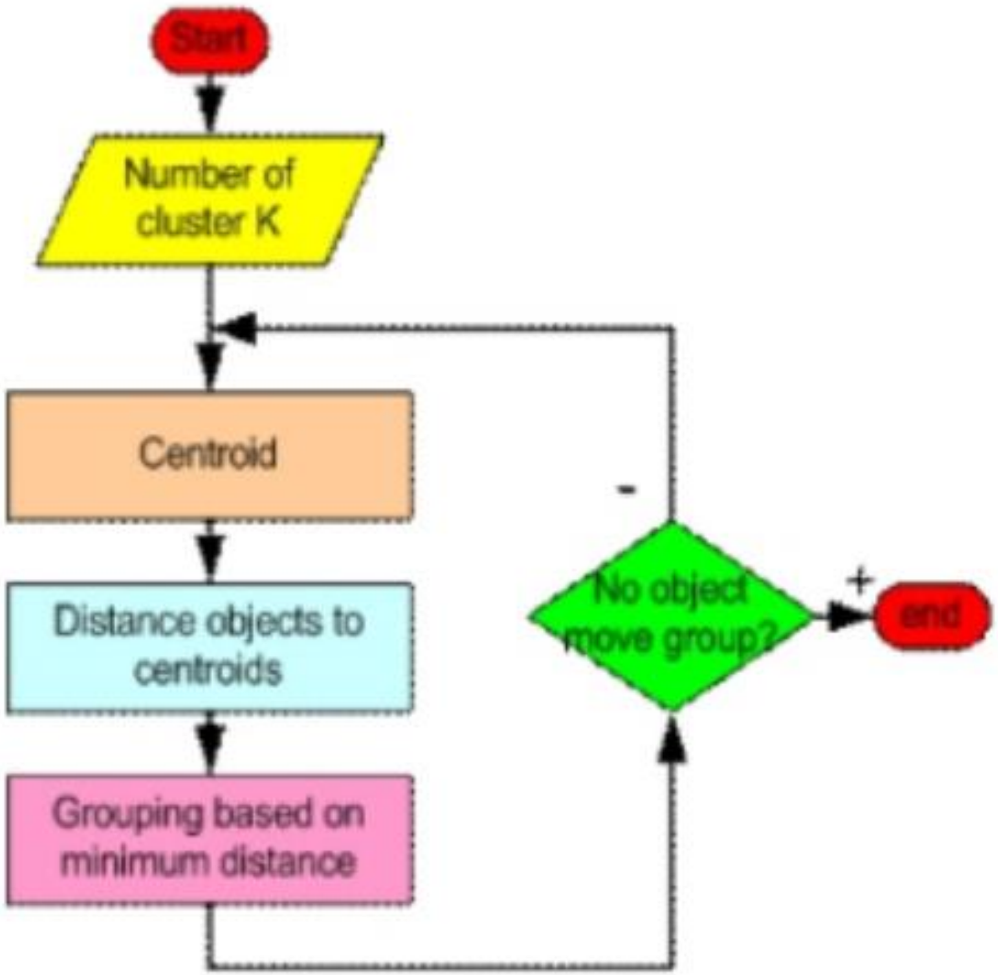
k-means clustering	k-means clustering is one of the simplest algorithms which uses unsupervised learning method to solve known clustering issues.
	Divides entire dataset into k clusters.
	k-means clustering require following two inputs. 1. K = number of clusters 2. Training set(m) = {x1, x2, x3,……, xm}





How does k-means clustering work?

- **Choose number of clusters**
- **Initialization**
- **Cluster assignment**
- **Move centroid**
- **Optimization**
- **Convergence**



k-means clustering: Step 1

- **Choose number of clusters**
- Initialization
- Cluster assignment
- Move centroid
- Optimization
- Convergence

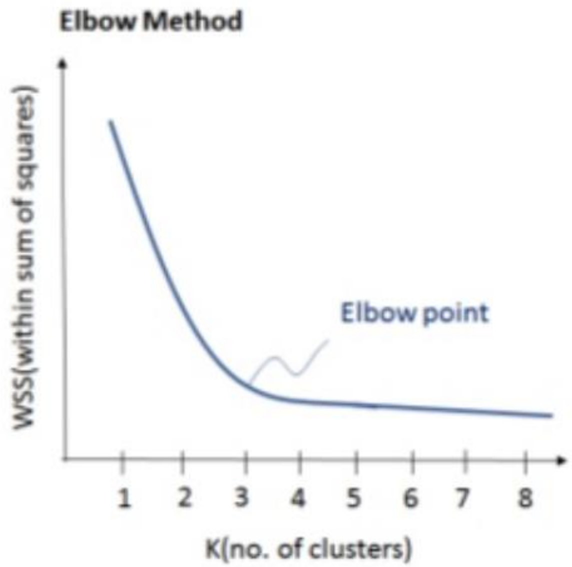
The WSS is defined as the sum of the squared distance between each member of the cluster and its centroid.

Mathematically:

$$WSS = \sum_{i=1}^{M} d(p_i, q^{(i)})^2 = \sum_{i=1}^{M} \sum_{j=1}^{n} \left(p_{ij} - q_j^{(i)}\right)^2$$

where, $p^{(i)}$= data point

$q^{(i)}$= closest centroid to data point



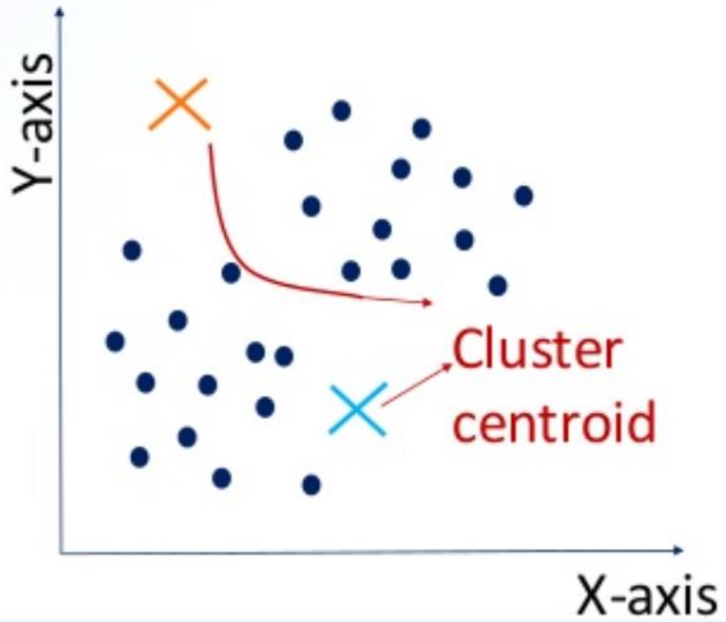
The idea of the elbow method is to choose the k after which the WSS decrease is almost constant.

k-means clustering: Step 2

- Choose number of clusters
- **Initialization**
- Cluster assignment
- Move centroid
- Optimization
- Convergence

➢ Randomly initialize *k* points called the cluster centroids.
Here, k = 2

➢ Value of k(number of clusters) can be determined by the elbow curve.

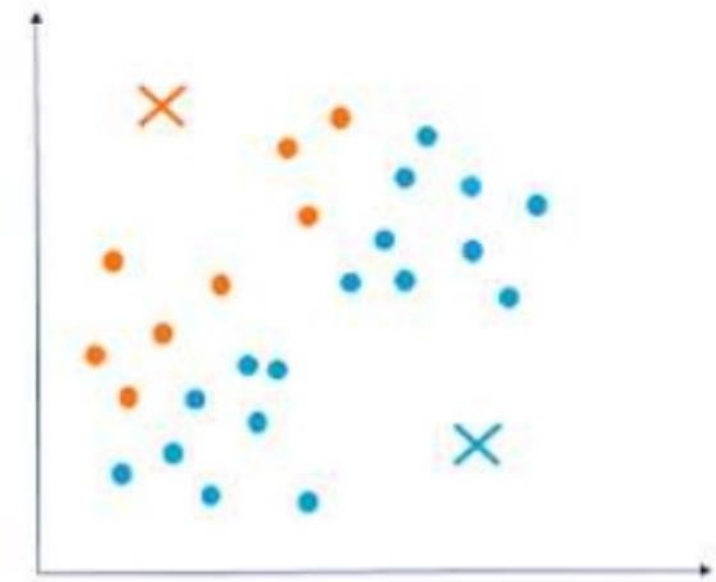


k-means clustering: Step 3

- Choose number of clusters
- Initialization
- **Cluster assignment**
- Move centroid
- Optimization
- Convergence

➢ Compute the distance between the data points and the cluster centroid initialized.

➢ Depending upon the minimum distance, data points are divided into two groups.



k-means clustering: Step 4

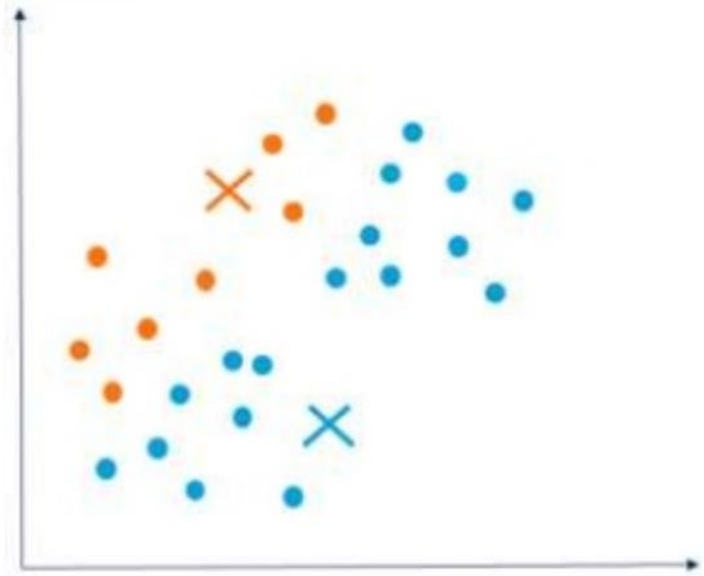
- Choose number of clusters
- Initialization
- Cluster assignment
- **Move centroid**
- Optimization
- Convergence

➢ Compute the mean of blue dots.

➢ Reposition blue cluster centroid to this mean.

➢ Compute the mean of orange dots.

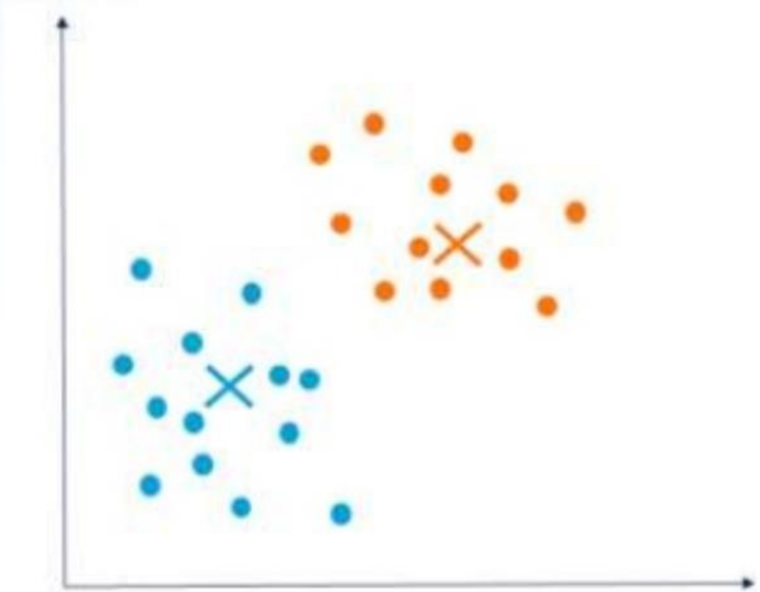
➢ Reposition orange cluster centroid to this mean.



k-means clustering: Step 5

- Choose number of clusters
- Initialization
- Cluster assignment
- Move centroid
- **Optimization**
- Convergence

➢ Repeat previous two steps iteratively till the cluster centroids stop changing their positions.

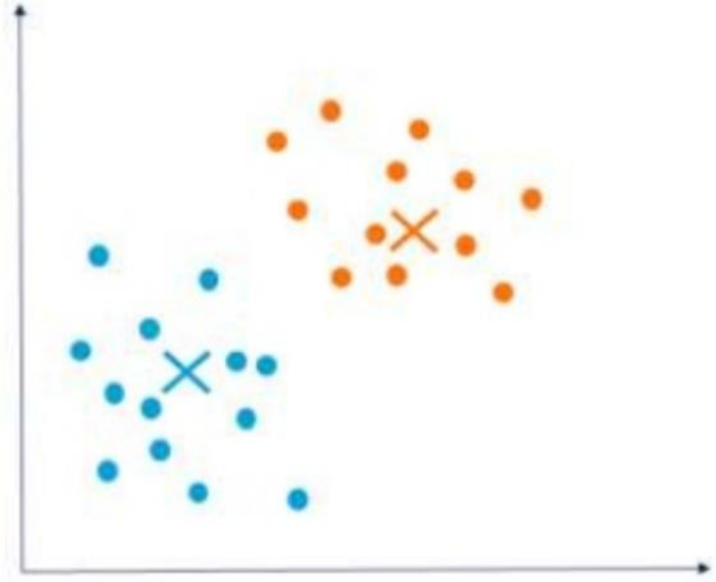


k-means clustering: Step 6

- Choose number of clusters
- Initialization
- Cluster assignment
- Move centroid
- Optimization
- **Convergence**

➢ Finally, k-means clustering algorithm converges.

➢ Divides the data points into two clusters clearly visible in orange and blue.



k-means clustering: Example

Dataset:

I	X1	X2
A	1	1
B	1	0
C	0	2
D	2	4
E	3	5

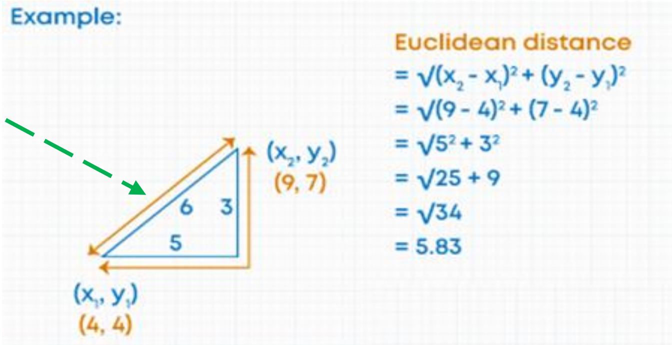
2 Clusters random Centroid:

	X1	X2	Centroid Value
C1	1	1	(1, 1)
C2	0	2	(0, 2)

Distance Calculation:

$$\text{Euclidean distance} = \sqrt{(\text{observe value} - \text{centroid value})^2 + (\text{observe value} - \text{centroid value})^2}$$

$$\text{Euclidean distance} = \sqrt{((X_x - X_1))^2 + (X_y - Y_1)^2}$$



Row 1 (A)

$C1 = \sqrt{(1-1)^2 + (1-1)^2} = 0$

$C2 = \sqrt{(1-0)^2 + (1-2)^2} = 1.4$

Row 2 (B)

$C1 = \sqrt{(1-1)^2 + (0-1)^2} = 1$

$C2 = \sqrt{(1-0)^2 + (0-2)^2} = 2.2$

Row 3 (C)

$C1 = \sqrt{(0-1)^2 + (2-1)^2} = 1.4$

$C2 = \sqrt{(0-0)^2 + (2-2)^2} = 0$

Row 4 (D)

$C1 = \sqrt{(2-1)^2 + (4-1)^2} = 3.2$

$C2 = \sqrt{(2-0)^2 + (4-2)^2} = 2.8$

Row 5 (E)

$C1 = \sqrt{(3-1)^2 + (5-1)^2} = 4.5$

$C2 = \sqrt{(3-0)^2 + (5-2)^2} = 4.2$

Now

Distance Matrix

I	C1=1	C2=2
A	0	1.4
B	1	2.2
C	1.4	0
D	3.2	2.8
E	4.5	4.2

I	C1=1	C2=2	Cluster
A	0	1.4	1
B	1	2.2	1
C	1.4	0	2
D	3.2	2.8	2
E	4.5	4.2	2

k-means clustering: Example

Dataset with Clusters:

I	X1	X2
A	1	1
B	1	0
C	0	2
D	2	4
E	3	5

Cluster 1

Cluster 2

I	X1	X2	Cluster
A	0.5	2.7	1
B	0.5	3.7	1
C	1.8	2.4	1
D	3.6	0.5	2
E	4.9	1.9	2

Calculate New Centroids:

$$X1 = \frac{1+1}{2} = 1 \ , \frac{1+0}{2} = 2$$

$$X1 = (1, 0.5)$$

$$X2 = \frac{0+2+3}{3} = 1.7 \ , \frac{2+4+5}{3} = 3.7$$

$$X2 = (1.7, 3.7)$$

So mean of Cluster I and 2

X1(Cluster 1) = (0.7 , 1)

X1(Cluster 2) = (2.5 , 4.5)

New Centroid

X1 = (0.7 , 1)

X2 = (2.5 , 4.5)

Previous Centroid

X1 = (1, 0.5)

X2 = (1.7,3.7)

New Centroid

X1 = (1, 0.5)

X2 = (1.7,3.7)

Previous Centroid

X1 = (1, 1)

X2 = (0, 2)

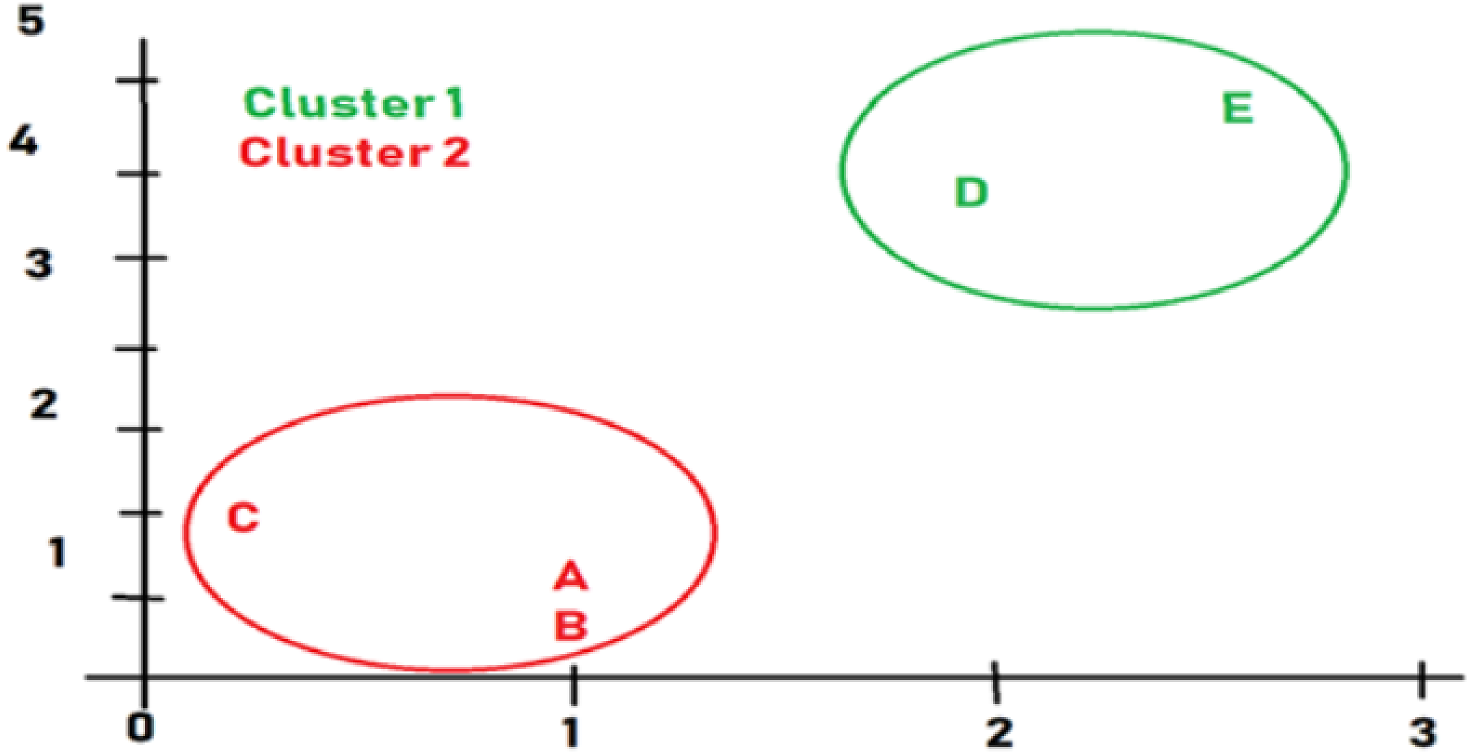
Distance Matrix With Respect to New Centroid

I	X1	X2	Cluster
A	0.5	2.7	1
B	0.5	3.7	1
C	1.8	2.4	1
D	3.6	0.5	2
E	4.9	1.9	2

k-means clustering: Example

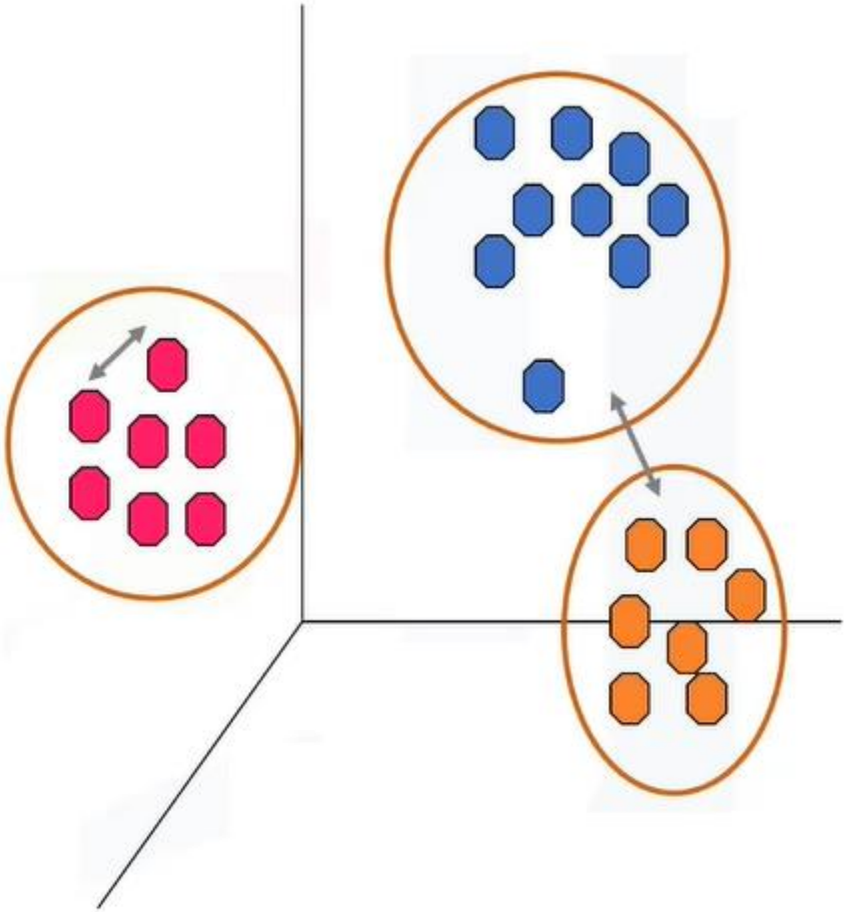
Keep on repeating until value of Centroids are equal

New Centroid	Previous Centroid
X1 = (1, 0.5)	X1 = (1, 0.5)
X2 = (1.7,3.7)	X2 = (1.7,3.7)

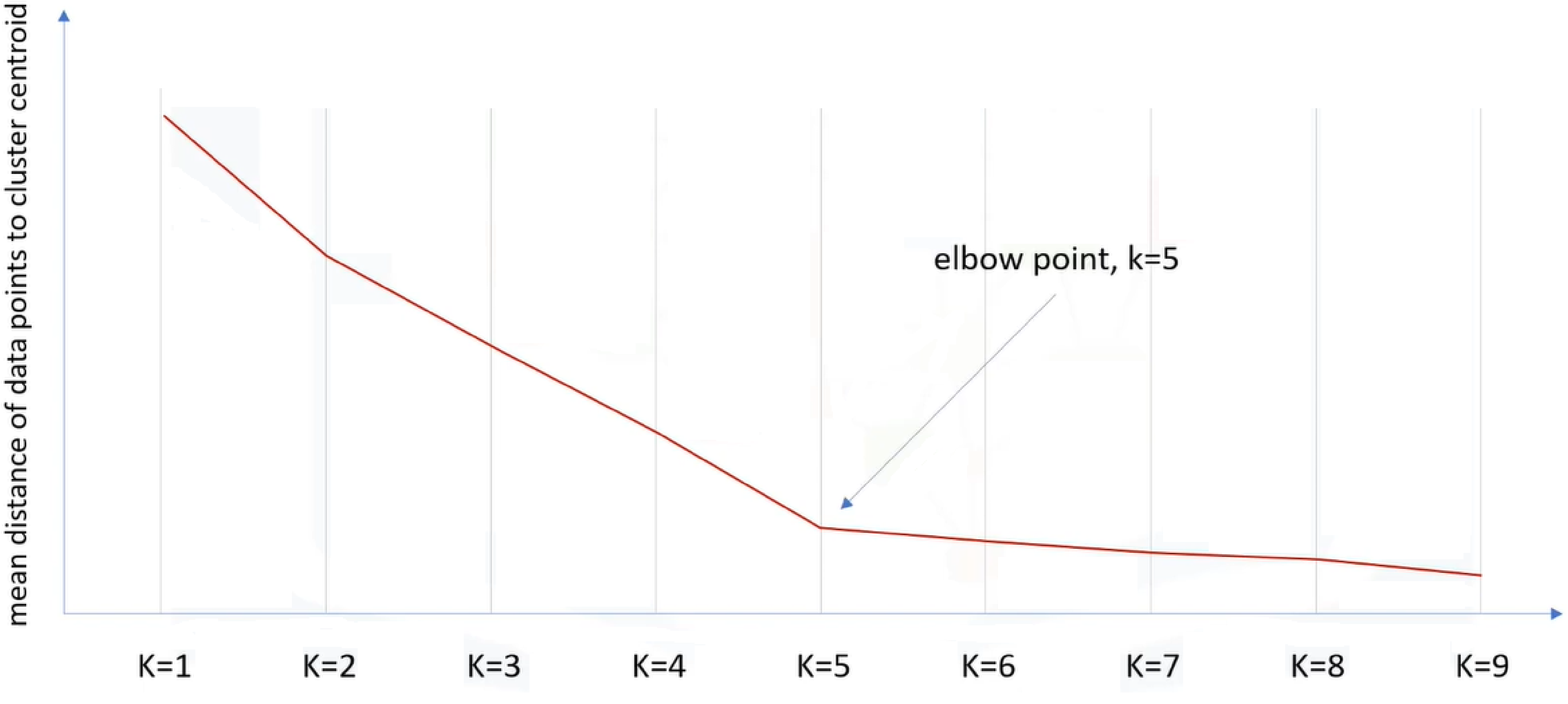


k-means clustering: Accuracy

- External approach
  - Compare the clusters with the ground truth, if it is available.
- Internal approach
  - Average the distance between data points within a cluster.



Choosing Value of k



k-means clustering: Review

1. Randomly placing *k* centroids, one for each cluster.
2. Calculate the distance of each point from each centroid.
3. Assign each data point (object) to its closest centroid, creating a cluster.
4. Recalculate the position of the *k* centroids.
5. Repeat the steps 2-4, until the centroids no longer move.

- Med and Large sized databases (*Relatively efficient*)
- Produces sphere-like clusters
- Needs number of clusters (k)

Advantages

1. K-means clustering is relatively simple to implement.

2. The algorithm is known to easily adapt to new examples.

3. It guarantees convergence by trying to minimize the total SSE as an objective function over a number of iterations.

4.The algorithm is fast and efficient in terms of computational cost.

Disadvantages

1. **Choosing** k **manually.** This is the greatest factor in the convergence of the algorithm and can provide widely different results for different values of k.

2. **Clustering outliers.** Outliers must be removed before clustering, or they may affect the position of the centroid or make a new cluster of their own.

3. **Being dependent on initial values.** As the value of k increases, other algorithms **(i.e. k-means seeding)** need to be applied to give better values for the initial centroids.

4. **Scaling with number of dimensions.** As the number of dimensions increases, the difficulty in getting the algorithm to converge increases due to the curse of dimensionality, discussed above.

Quiz

What is the minimum no. of variables/ features required to perform clustering?

A. 0

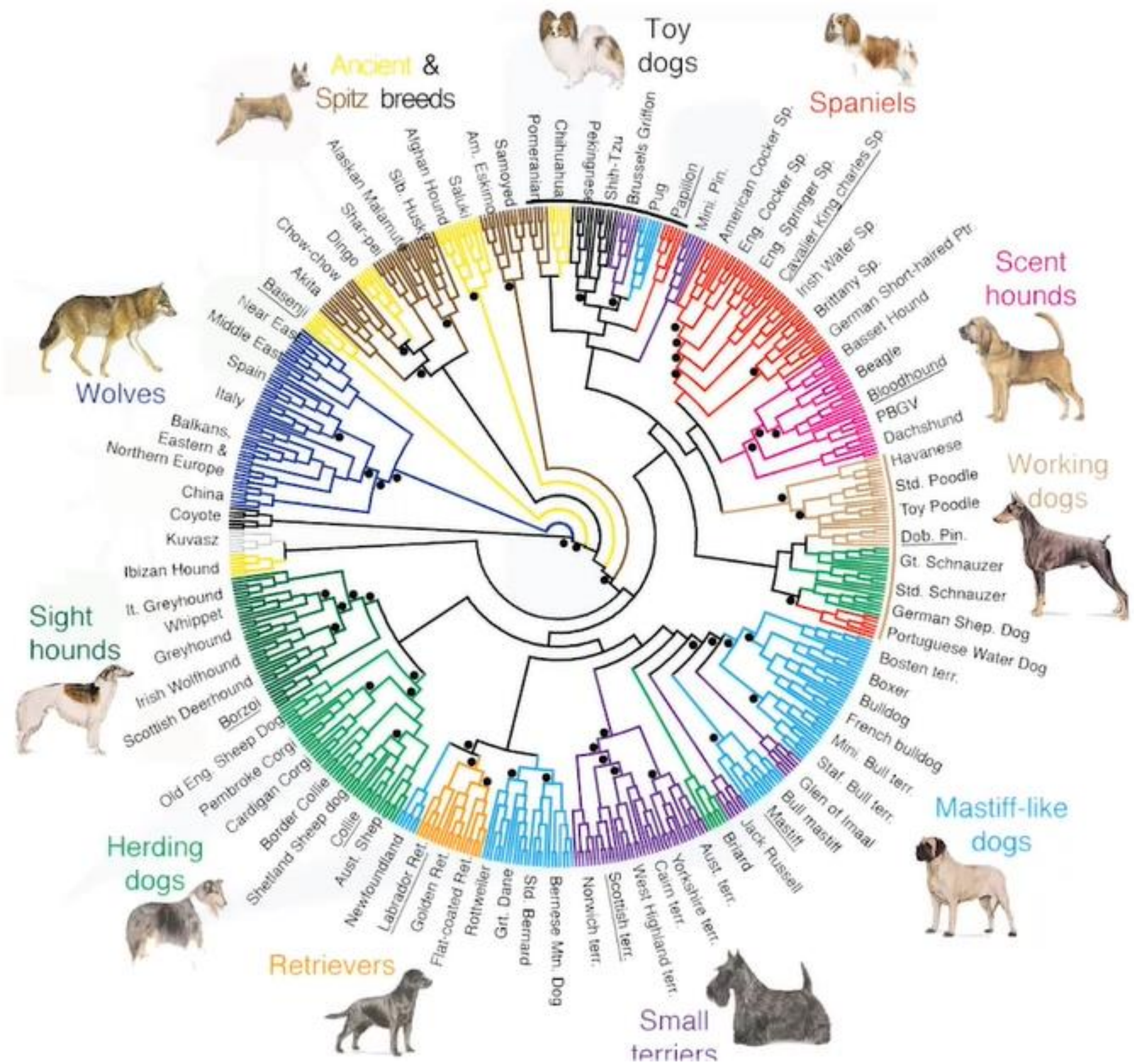
B. 1

C. 2

D. 3

Hierarchial Clustering

Hierarchical Clustering

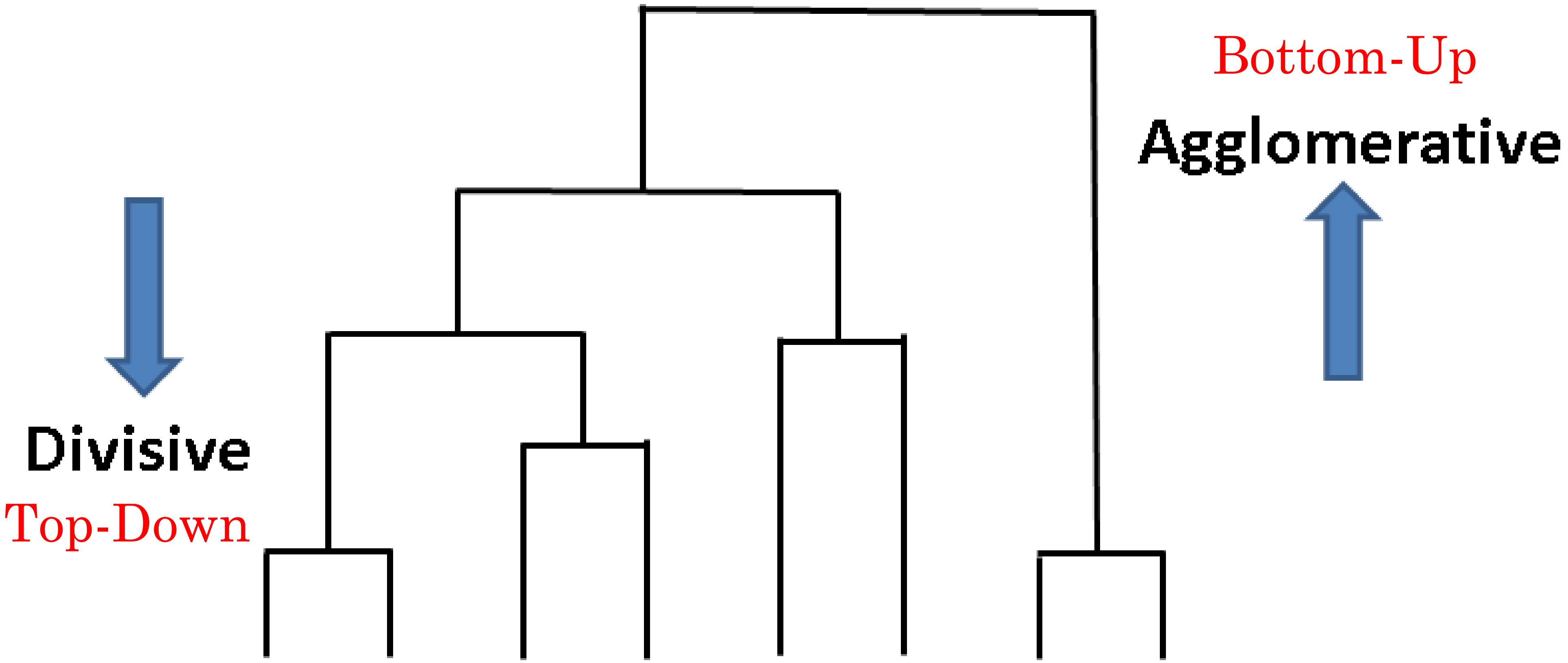


Hierarchical clustering algorithms build a hierarchy of clusters where each node is a cluster consists of the clusters of its daughter nodes.

Source: von Holdt B.M. et al. (2010) Genome-wide SNP and haplotype analyses

Hierarchical Clustering

- ***Hierarchical clustering***, also known as *hierarchical cluster analysis,* is an algorithm that groups similar objects into groups called *clusters*. The endpoint is a set of clusters, where each cluster is distinct from each other cluster, and the objects within each cluster are broadly similar to each other.

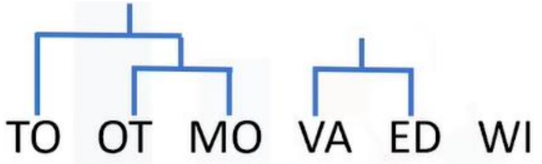


Agglomerative Hierarchical Clustering

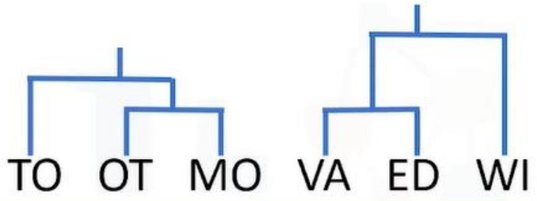
	TO	OT	VA	MO	WI	ED
TO		351	3363	505	1510	2699
OT			3543	167	1676	2840
VA				3690	1867	819
MO					1824	2976
WI						1195
ED						

	TO	OT/MO	VA	WI	ED
TO		351	3363	1510	2699
OT/MO			3543	1676	2840
VA				1867	819
WI					1195
ED					

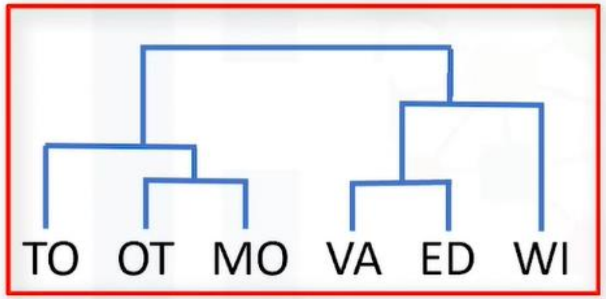
Agglomerative Hierarchical Clustering



	TO/OT/MO	VA	WI	ED
TO/OT/MO		3543	1676	2840
VA			1867	819
WI				1195
ED				



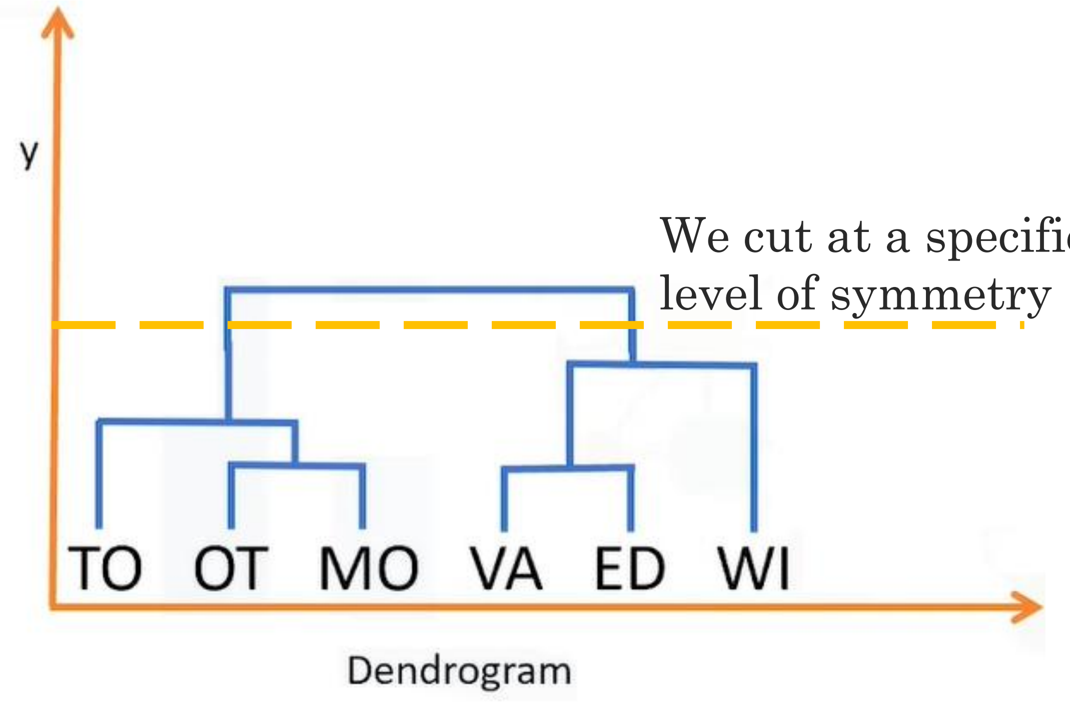
	TO/OT/MO	VA/ED	WI
TO/OT/MO		2840	1676
VA/ED			1667
WI			



	TO/OT/MO	VA/ED/WI
TO/OT/MO		1676
VA/ED/WI		

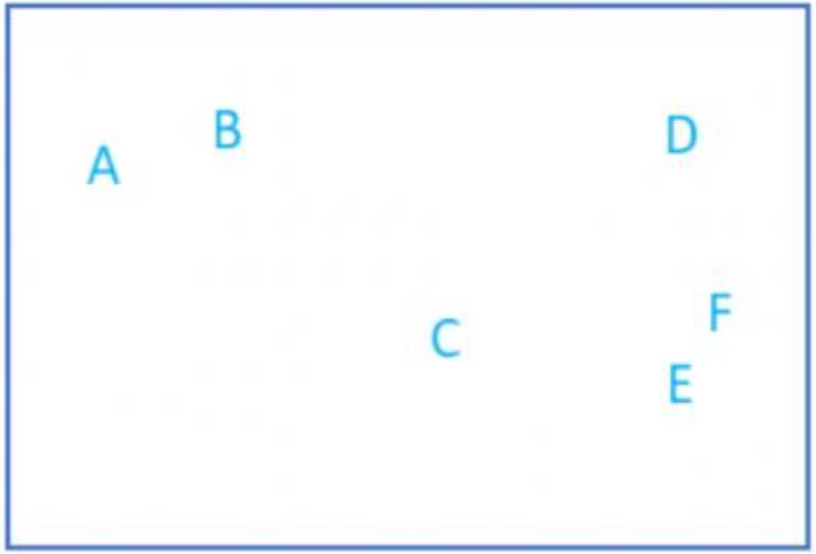
Agglomerative Hierarchical Clustering

We cut at a specific level of symmetry



Agglomerative Hierarchical Clustering

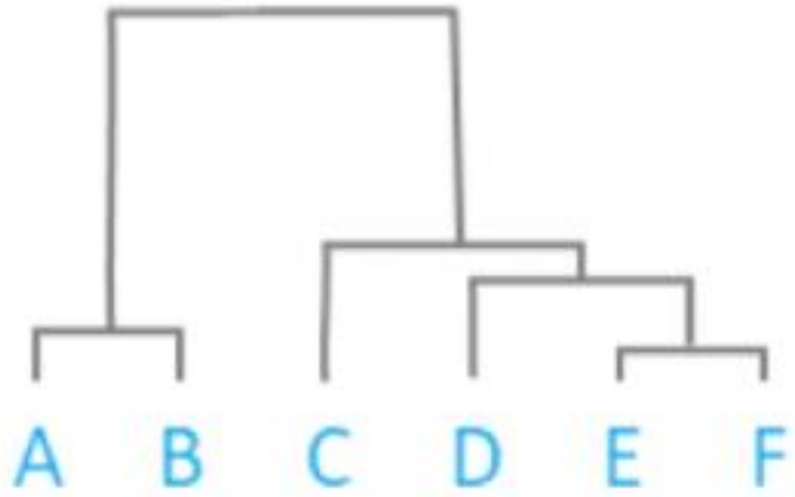
Raw Data



Distance Matrix

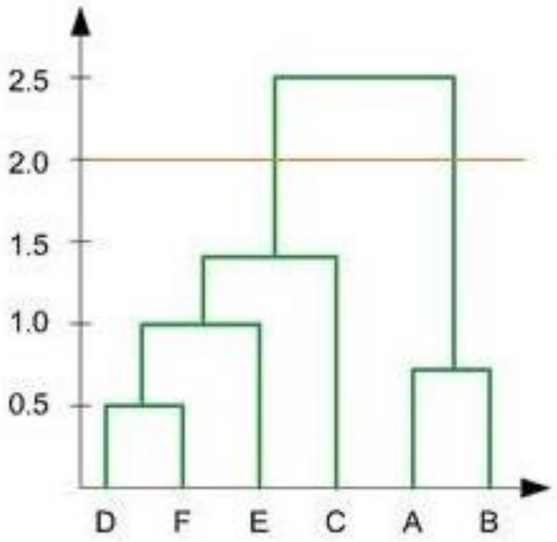
	A	B	C	D	E
B	16				
C	47	37			
D	72	57	40		
E	77	65	30	31	
F	79	66	35	23	10

Dendrogram



Quiz

In the figure below, if you draw a horizontal line on y-axis for y=2. What will be the number of clusters formed?



A. 1

B. 2

C. 3

D. 4

Heirarchical Algorithm

Advantages	Disadvantages
Doesn't required number of clusters to be specified.	Can never undo any previous steps throughout the algorithm.
Easy to implement.	Generally has long runtimes.
Produces a dendrogram, which helps with understanding the data.	Sometimes difficult to identify the number of clusters by the dendrogram.

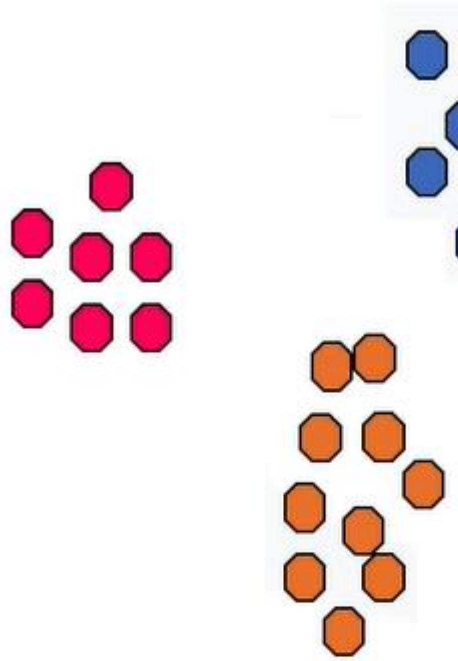
K-means vs. Heirarchical Clustering

K-means	Hierarchical Clustering
1. Much more efficient	1. Can be slow for large datasets
2. Requires the number of clusters to be specified	2. Does not require the number of clusters to run
3. Gives only one partitioning of the data based on the predefined number of clusters	3. Gives more than one partitioning depending on the resolution
4. Potentially returns different clusters each time it is run due to random initialization of centroids	4. Always generates the same clusters

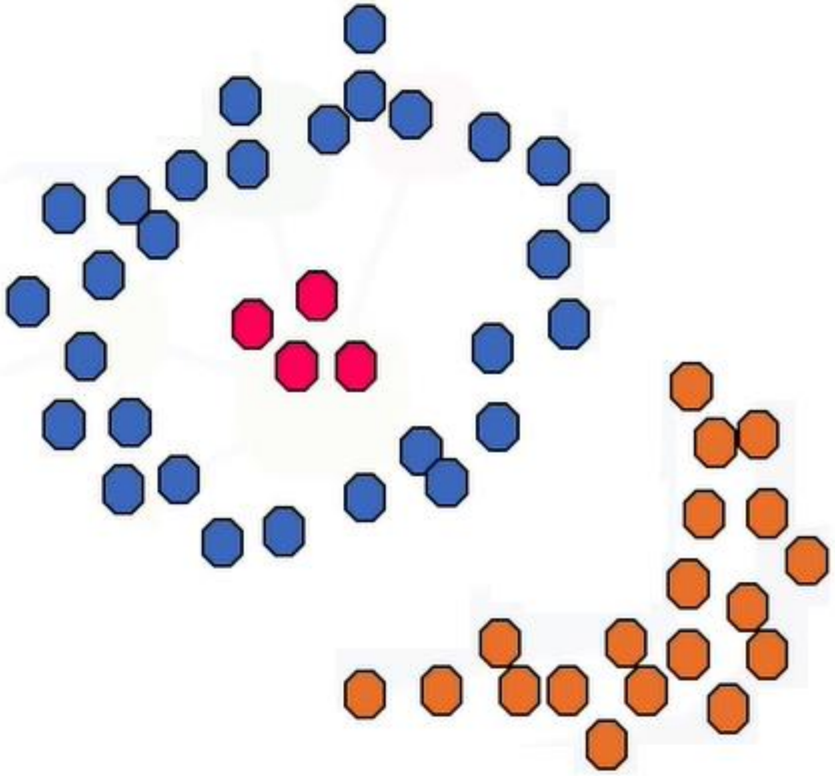
Density Based Clustering

Density Based Clustering

- Spherical-shape clusters



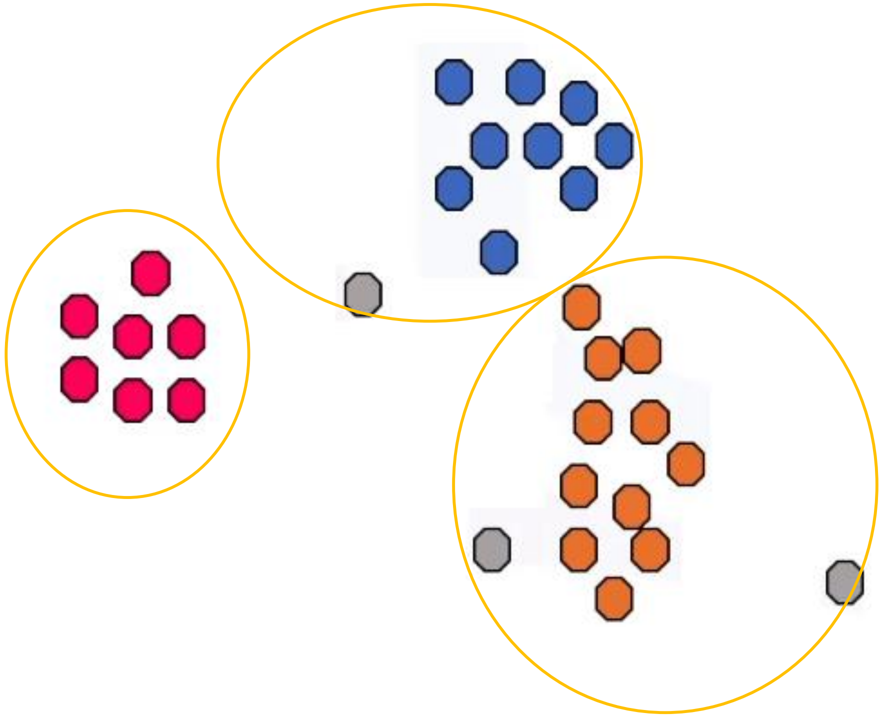
- Arbitrary-shape clusters

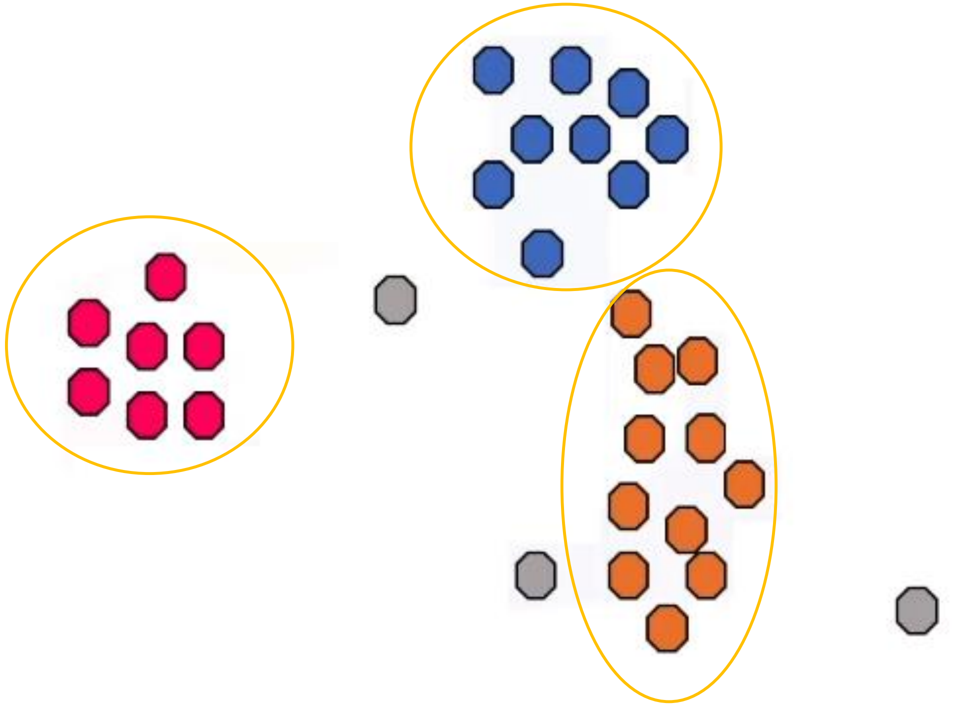


K-Means vs. Density Based Clustering

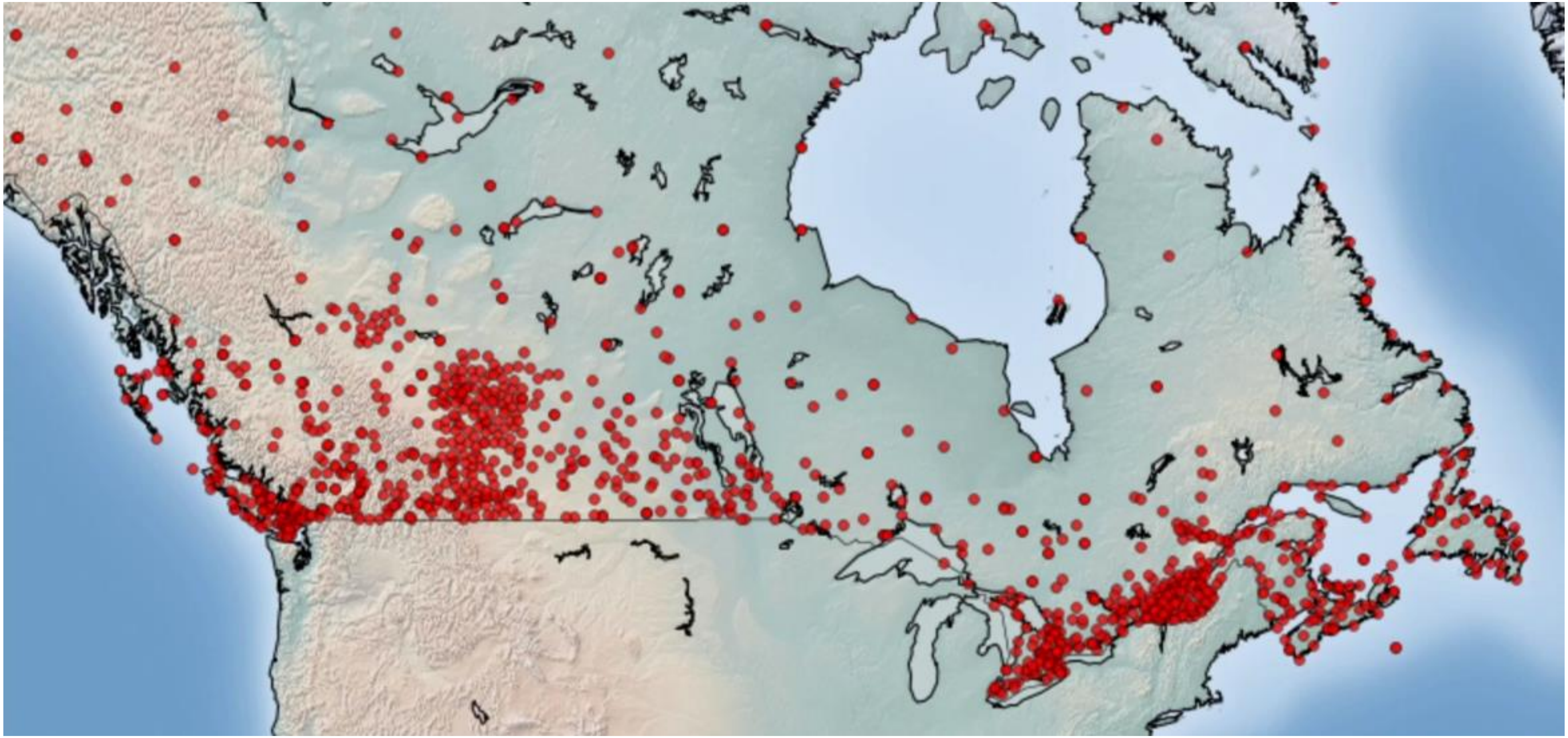
- k-Means assigns all points to a cluster even if they do not belong in any

- Density-based Clustering locates regions of high density, and separates outliers



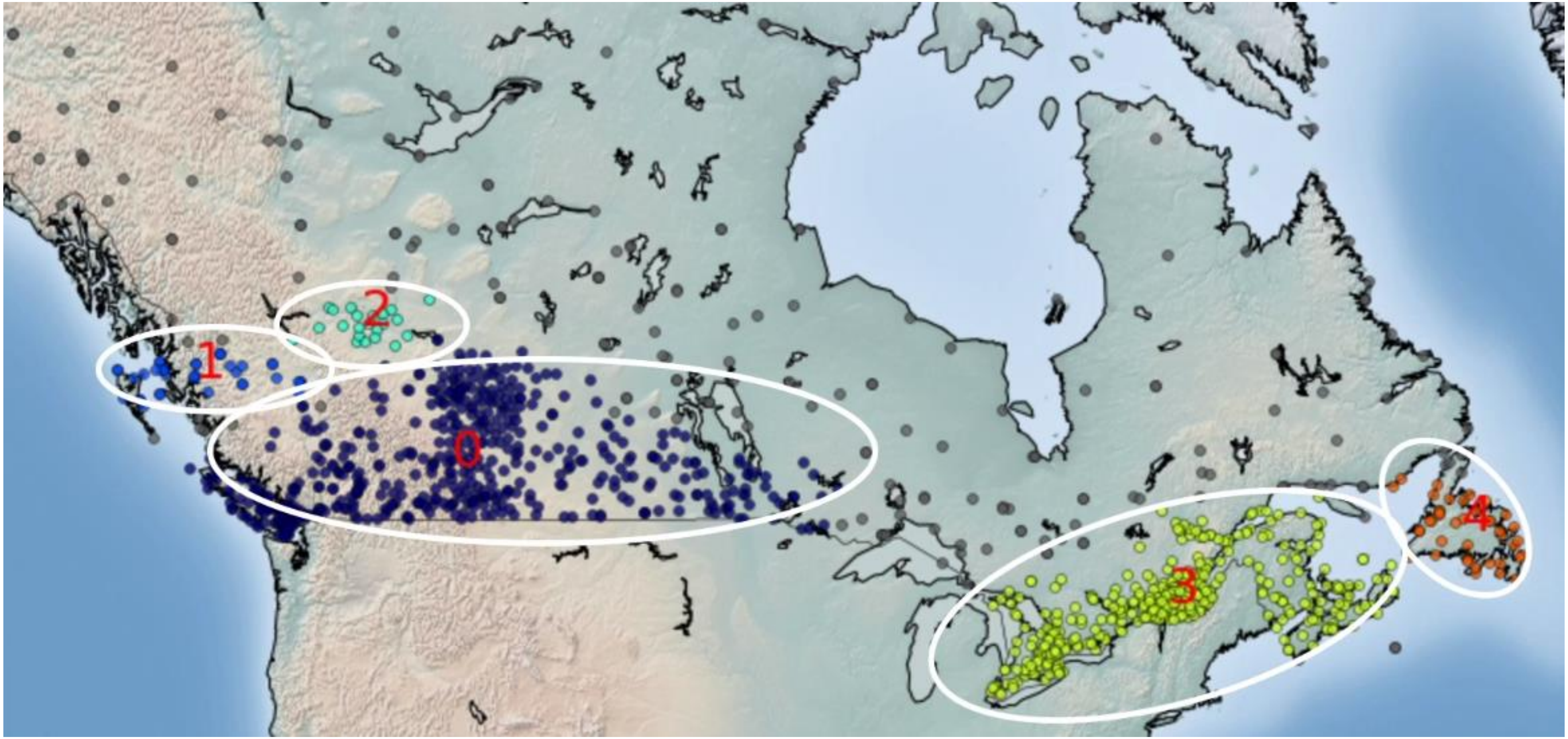


DBSCAN for Class Identification



Dots Represent Weather Stations in Canada

DBSCAN for Class Identification



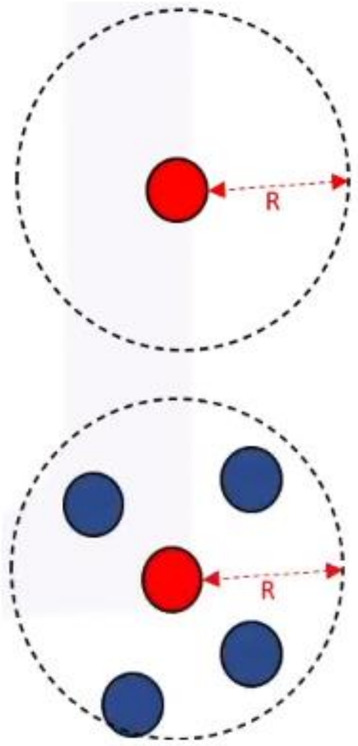
Stations Showing same Weather Condition Are Grouped in Clusters

What is DBSCAN?

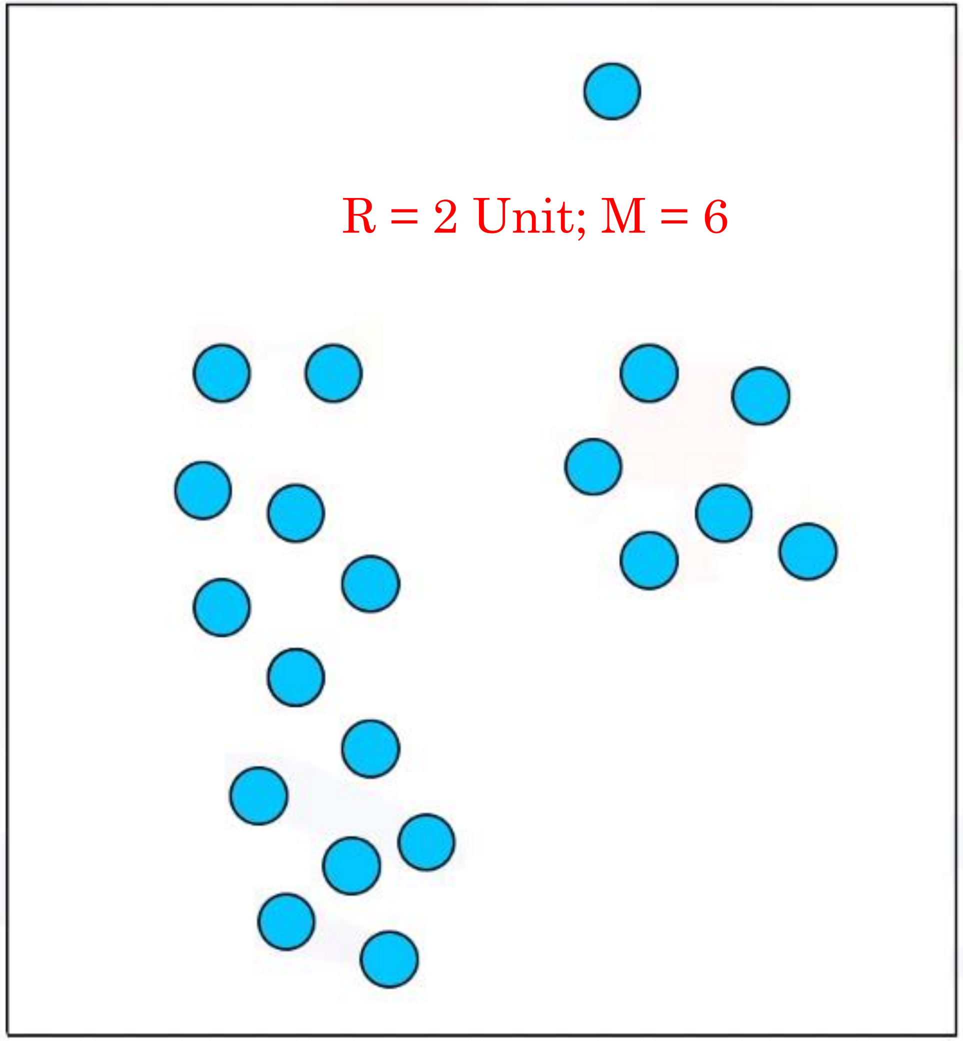
- DBSCAN (**D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with **N**oise)
  - Is one of the most common clustering algorithms
  - Works based on density of objects

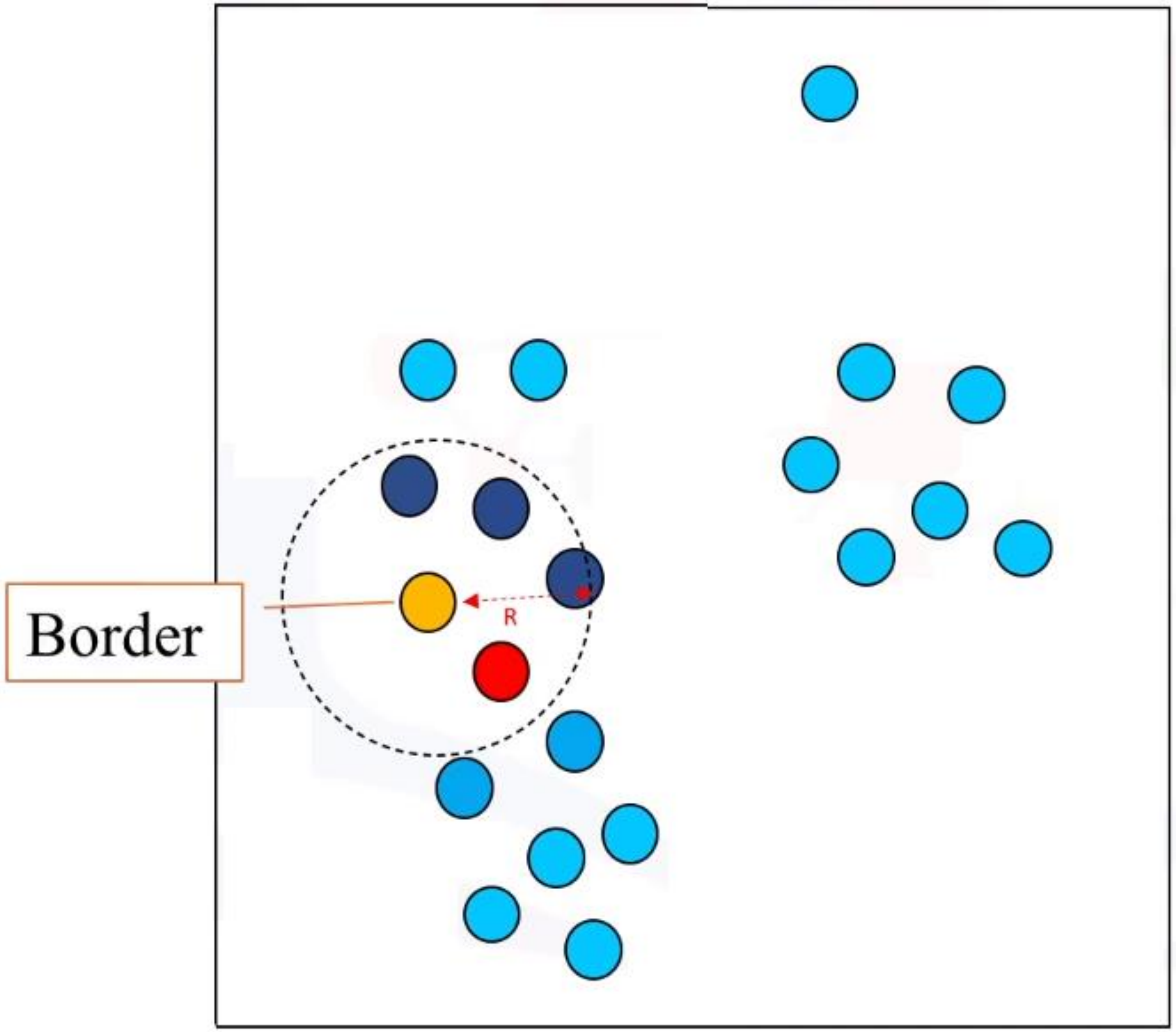
- R (**R**adius of neighborhood)
  - Radius (R) that if includes enough number of points within, we call it a dense area

- M (**M**in number of neighbors)
  - The minimum number of data points we want in a neighborhood to define a cluster

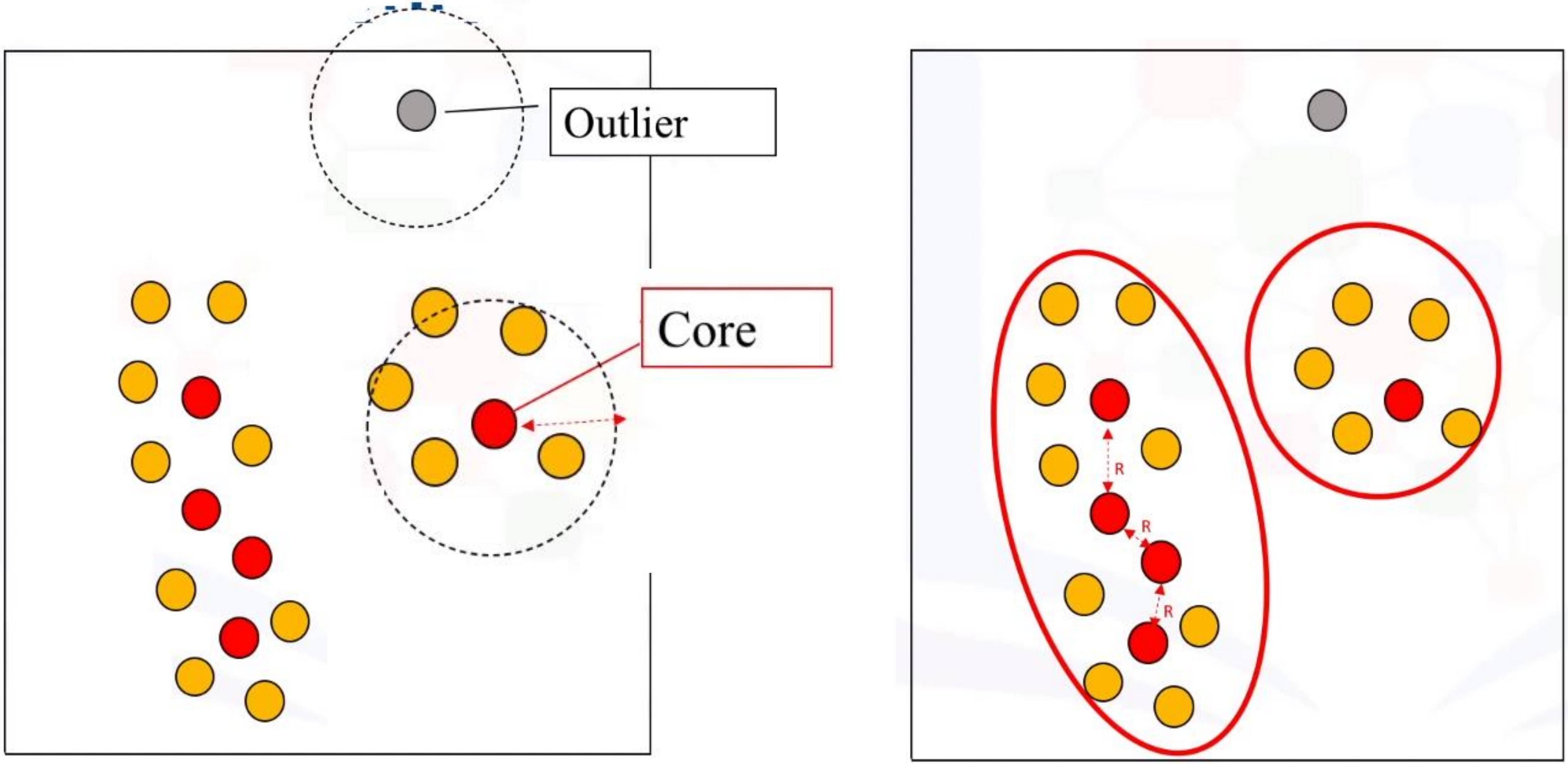


How DBSCAN Works?



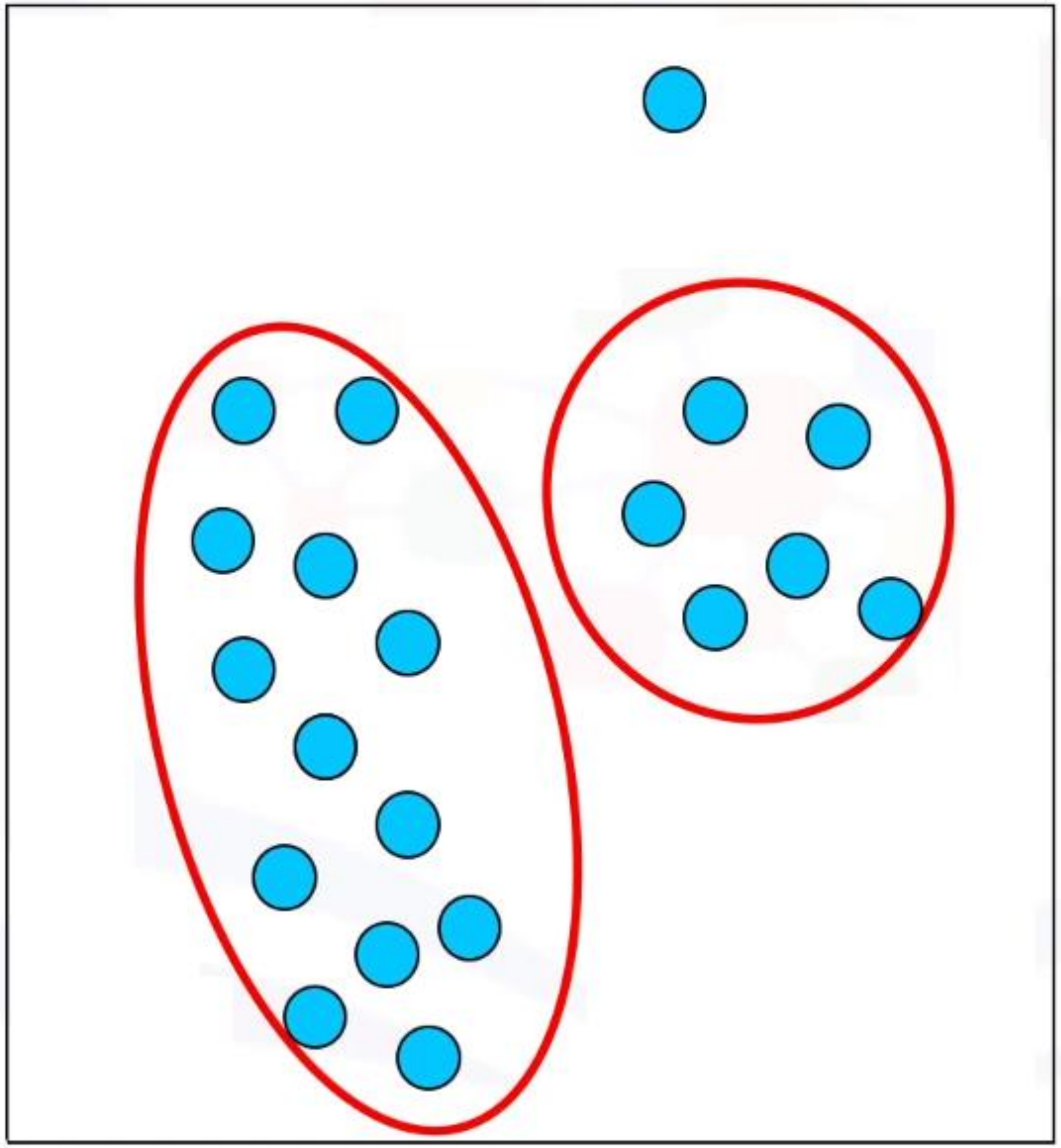


How DBSCAN Works?



Demo: https://www.naftaliharris.com/blog/visualizing-dbscan-clustering/

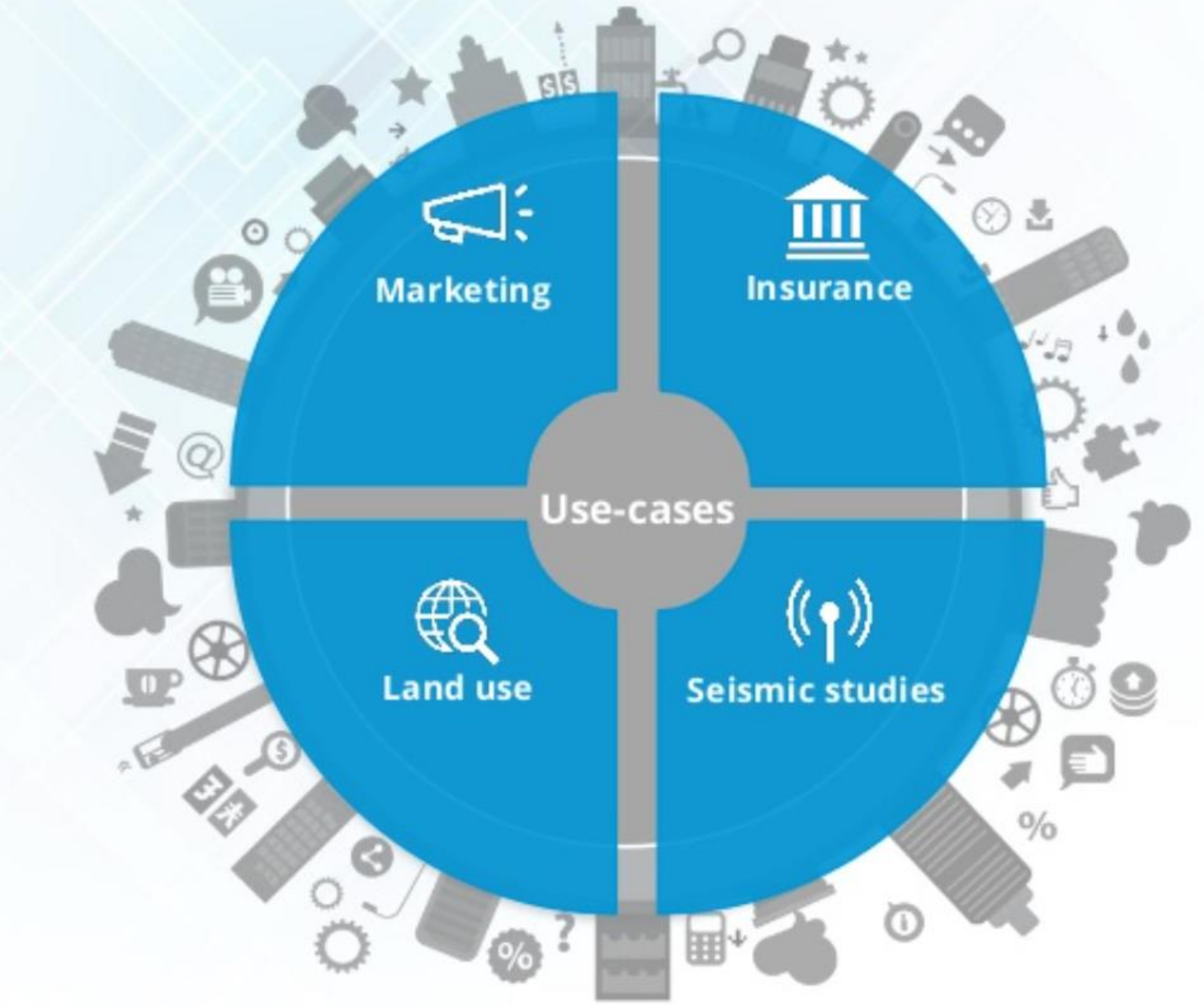
Advantages of DBSCAN



1. Arbitrarily shaped clusters
2. Robust to outliers
3. Does not require specification of the number of clusters

Clustering Applications

Clustering Application



Marketing
Discovering distinct groups in customer databases, such as customers who make lot of long-distance calls.

Insurance
Identifying groups of crop insurance policy holders with a high average claim rate. Farmers crash crops, when it is “profitable”.

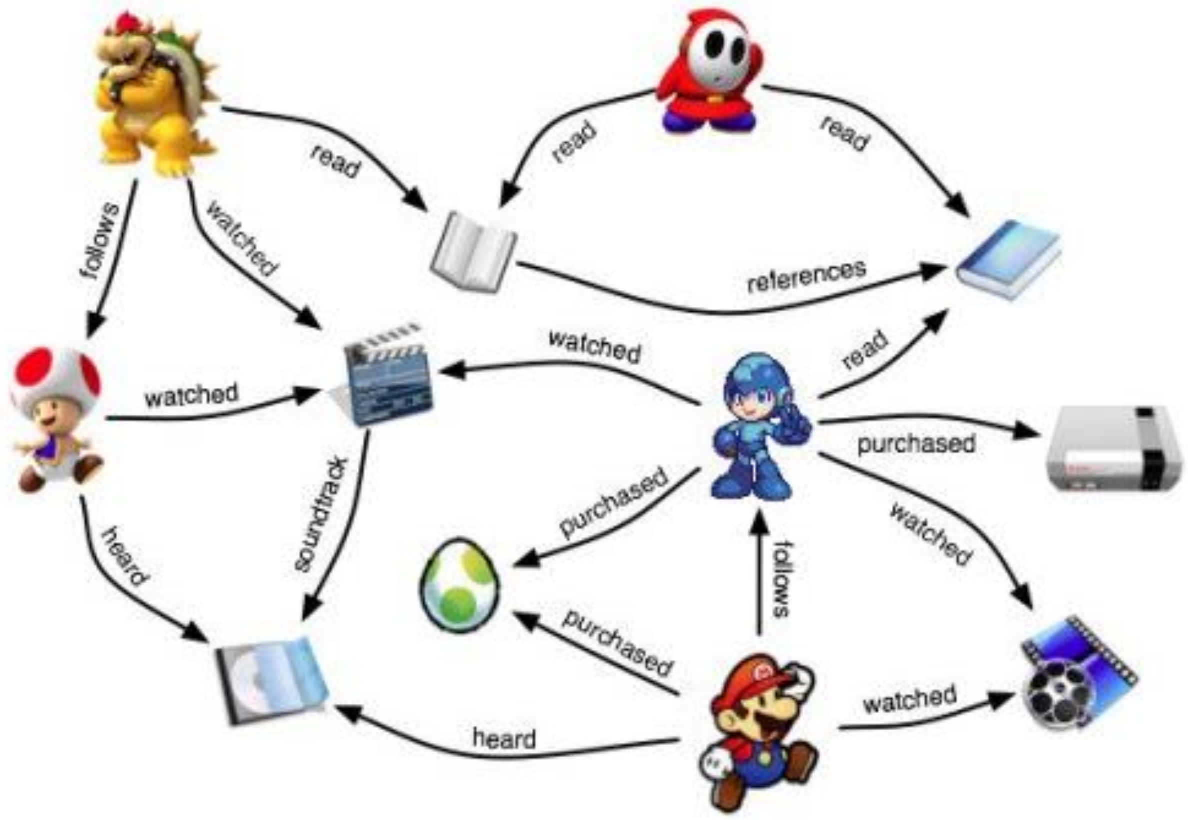
Land use
Identification of areas of similar land use in a GIS database.

Seismic studies
Identifying probable areas for oil/gas exploration based on seismic data

Clustering Application

Recommender Systems

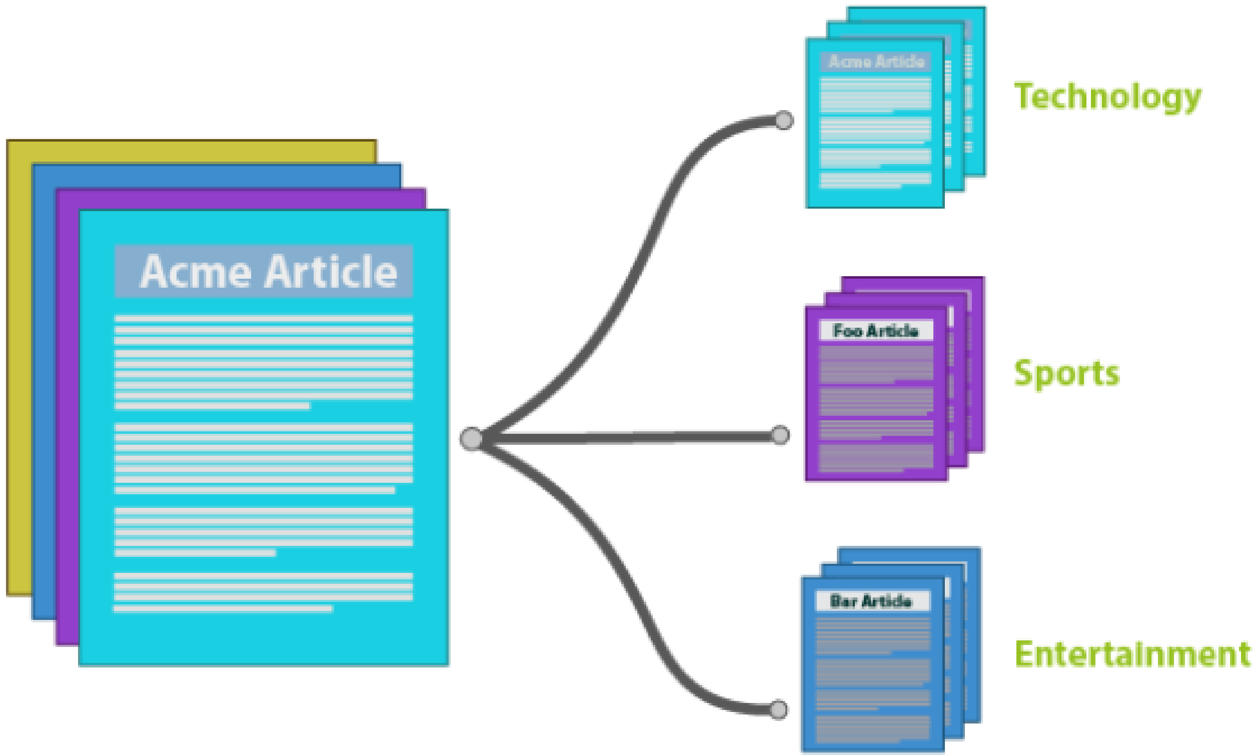
Clustering can also be used in recommendation engines. In the case of recommending movies to someone, you can look at the movies enjoyed by a user and then use clustering to find similar movies.



Clustering Application

Document Clustering

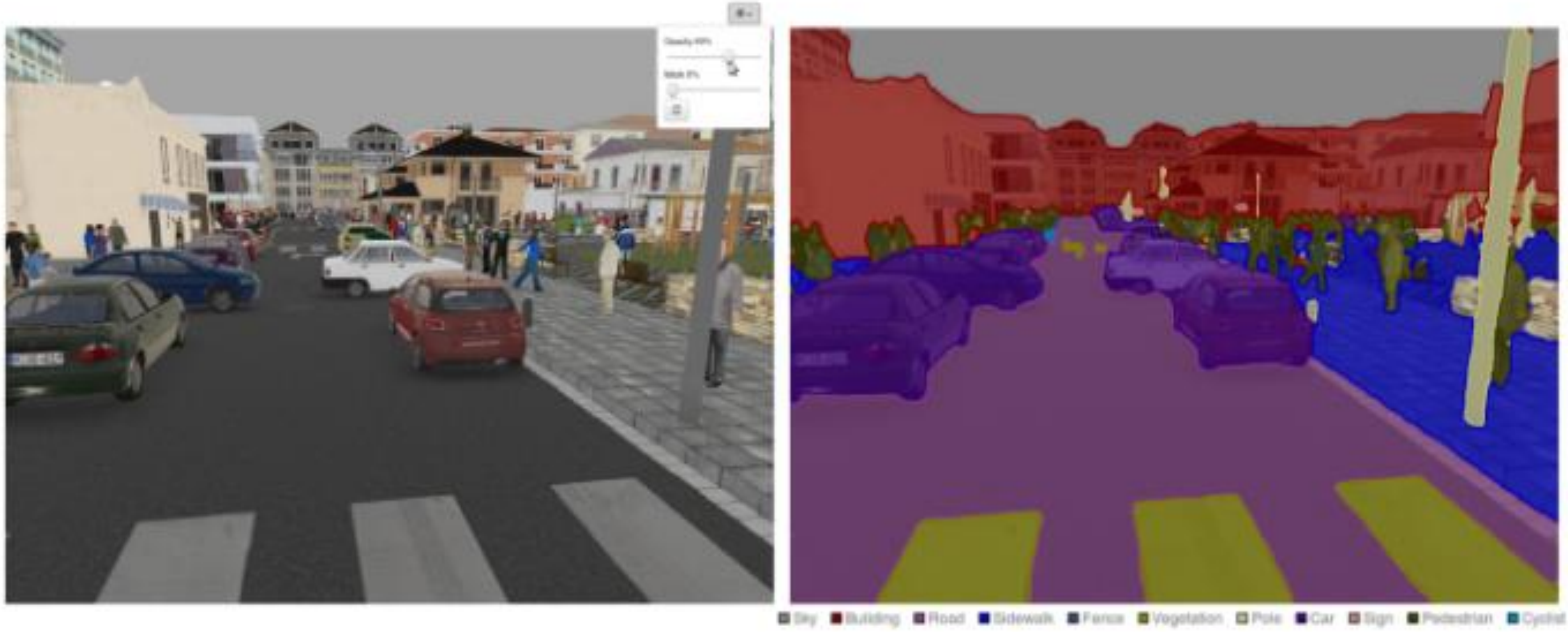
This is another common application of clustering. Let's say you have multiple documents and you need to cluster similar documents together. Clustering helps us group these documents such that similar documents are in the same clusters.



Clustering Application

Image Segmentation

Image segmentation is a wide-spread application of clustering. Similar pixels in the image are grouped together. We can apply this technique to create clusters having similar pixels in the same group.



Quiz

Q1. Movie Recommendation systems are an example of:

1. **Classification**
2. **Clustering**
3. **Reinforcement Learning**
4. **Regression**

Options:

B. A. 2 Only

C. 1 and 2

D. 1 and 3

E. 2 and 3

F. 1, 2 and 3

H. 1, 2, 3 and 4

Thank You for Attention